MP aes attributes backends basexx bit32 blobs
boolean buffers builders caches callback
callbacks catcodes characters chemistry
commands containers context converters
coroutine ctxrunner debug dir directives
document documentdata drivers environment
experiments figures file fio floats font fonts
functions global graphics gzip images
interactions interfaces io job kpse language
languages layouts lfs libraries lmx loaders
logs lpdf lpeg lua luac luatex lxml math
mathematics md5 metapost mime moduledata
modules mp mplib noads node nodes number
objects optional os package pagebuilders
parametersets pdf pdfdecode pdfe physics
pngdecode posit potrace publications regimes
resolvers sandbox scripts set sha2 sio socket
sorters sparse statistics status storage
streams string structures table tex
texconditionals texconfig texconstants texifs
texio texisdefined texmodes texsystemmodes
thirddata token tokens tracers trackers
typesetters updaters url userdata utf8
utilities visualizers xcomplex xdecimal xmath
xml xzip zip zlib

ConTEXt LMTX

global (alias: _G)

CONTEXTLMTXMODE	number
LUAMAJORVERSION	number
LUAMINORVERSION	number
LUATEXENGINE	string
LUATEXFORMATID	number
LUATEXFUNCTIONALITY	number
LUATEXVERBOSE	string
LUATEXVERSION	number
LUAVERSION	number
MP	function
aes	table
arg	table
asnumber_p	function
assert	function
attributes	table
backends	table
basexx	table
bit32	table
blobs	table
blockrequire	function
boolean	table
buffers	table
builders	table
caches	table
callback	table
callbacks	table
catcodes	table
characters	table
checkimage	function
chemistry	table
collectgarbage	function
collectstructure	function
combine	function
commands	table
containers	table
context	function
converters	table
copyarray	function
copydictionary	function
coroutine	table
count	number
createmask	function
ctxrunner	table
debug	table
dimen	function
dir	table
directives	table
document	table
documentdata	table
dofile	function
drivers	table
environment	table
error	function
experiments	table
figures	table
file	table
fio	table
floats	table
flushsimplerule	function
flushspecialrule	function
font	table
fonts	table
functions	table
getmetatable	function
getscriptstatus	function
global	table
graphics	table
gzip	table
handledefinitions	function
handletransformstring	function
images	table
inspect	function
interactions	table
interfaces	table
io	table
ipairs	function
job	table
kpse	table
language	table
languages	table
last_pdf	number
layouts	table

lfs	table
libraries	table
lmx	table
load	function
loaders	data table
loadfile	function
loadstring	function
logs	table
lpdf	table
lpeg	table
lua	table
luac	table
luatex	table
lxml	table
math	table
mathematics	table
mathshapekern	function
md5	table
metapost	table
mime	table
moduledata	table
modules	table
mp	function
mplib	table
next	function
noads	table
node	table
nodes	table
number	table
objects	table
optional	table
optionalrequire	function
os	table
p_number_p	userdata
p_number_u	userdata
package	table
pagebuilders	table
pairs	function
parametersets	table
pcall	function
pdf	table
pdfdecode	table
pdfe	table
physics	table
pngdecode	table
popleaderlevel	function
posit	table
potrace	table
potracefrompng	function
print	function
profilelines	function
publications	table
pushleaderlevel	function
rawequal	function
rawget	function
rawlen	function
rawset	function
recolor	function
regimes	table
require	function
resolvers	table
ruledspace	function
sandbox	table
scripts	table
select	function
set	table
setinspector	function
setmathparameters	function
setmetatable	function
sha2	table
shipbox_ref_h	number
shipbox_ref_v	number
sio	table
socket	table
sorters	table
sparse	table
statistics	table
status	table
storage	table
streams	table
string	table
structures	table
table	table
tex	table
texconditionals	data table
texconfig	table
texconstants	data table
texifs	data table

texio	table	**url**	table
texisdefined	data table	**userdata**	table
texmodes	data table	**utf**	table
texsystemmodes	data table	**utf8**	table
thirddata	table	**utilities**	table
toboolean	function	**visualizers**	table
todecimal	function	**warn**	function
token	table	**width**	number
tokens	table	**xcomplex**	table
tonumber	function	**xdecimal**	table
tostring	function	**xmath**	table
tracers	table	**xml**	table
trackers	table	**xpcall**	function
type	function	**xshift**	number
typesetters	table	**xzip**	table
unpack	function	**zip**	table
updaters	table	**zlib**	table

aes (extra lua)

decode function

encode function

random function

attributes (context)

alternates table
- N attribute
- D data
- F enable
- F handler
- F register
- D registered
- D values

applyoverloads function

checkvalues function

colorintents table
- N attribute
- D data
- F enable
- F handler
- F register
- D registered

colors table
- N attribute
- F cmyk
- F cmyktogray
- F cmyktorgb
- F colorcomponents
- D counts
- B couple
- F currentmodel
- F currentnamedmodel
- D data
- N default
- F defineintermediatecolor
- F definemultitonecolor
- F defineprocesscolor
- F defineprocesscolordirect
- F definesimplegray
- F definespotcolor
- F definetransparency
- F enable
- F filter
- N forced
- F forcedmodel
- F forcesupport
- F formatcolor
- F formatgray
- F formatluminance
- F formatluminanceratio
- F getpagecolormodel
- F gray
- F graytohsv
- F handler
- T helpers
  - F complement
  - F inbetween
  - F justone
- F hsvtogray
- F hsvtorgb
- F hwbtorgb
- F iccprofile
- F isblack
- F labtorgb
- F labtoxyz
- N main
- S model
- D models
- F mpcolor
- F mpnamedcolor
- F mpoptions
- F namedcolorattributes
- N none
- F popset
- F processcolorcomponents
- F pushset
- F register
- D registered
- F registerspotcolor
- F rgb
- F rgbtocmyk
- F rgbtogray
- F rgbtohsv
- F rgbtoxyz
- N selector
- F setlist
- F setmodel

F setpagecolormodel
D sets
F spec
F spot
F spotcolorname
F spotcolorparent
F spotcolorvalue
B supported
F toattributes
F transparencycomponents
B triggering
F usecolors
D valid
F value
D values
B weightgray
F xyztolab
F xyztorgb

counts data table
define function
effects table
N attribute
D data
F enable
F handler
F register
D registered
D values
getvalue function
getvalues function
handlers table
hasvalues function
list data table
markasoverload function
names data table
negatives table
N attribute
D data
F enable
F handler
F register
D registered
F set
numbers data table
ofnode function
private function
public function
registervalue function
restore function
save function
setcleaner function
showcurrent function
states table
F disabletriggering
F enabletriggering
F finalize
F initialize
F process
F selective
F simple
F stacked
F stacker
system function
tooverloads function
transparencies table
N attribute
D data
F enable
F forcesupport
F handler
T names
N color
N colorburn
N colordodge
N darken
N difference
N exclusion
N hardlight
N hue
N lighten
N luminosity
N multiply
N none
N normal
N overlay
N saturation
N screen
N softlight

N none
F register
D registered
B supported
F toattribute
B triggering
F value
D values

unsetvalue number

values data table

viewerlayers table

N attribute
F cleanup
D data
F define
F definedlayoutcomponent
F enable
F forcesupport
F handler
B hasorder
D listwise
F register
D registered
F resolve_begin
F resolve_end
F resolve_reset
F resolve_step
D scopes
F setfeatures
F setup
F start
F stop
B supported
D values

backends (context)

closeout function
codeinjections table
- F registerboxresource
- F registerfont
- F restoreboxresource

current string
defaults table
- T codeinjections
  - F prerollreference
  - F setautoprefix
  - F vfstartcolor
  - F vfstopcolor
- T nodeinjections
  - F cmykcolor
  - F destination
  - F graycolor
  - F injectknockout
  - F injectoverprint
  - F reference
  - F rgbcolor
  - F spotcolor
  - F transparency
- T registrations
  - F transparency
- D tables

getcallbackstate function
included table
- B context
- B date
- B id
- B metadata
- B pdf

initialize function
latelua function
nodeinjections table
nothing function
openout function
registered table
- T pdf
  - T codeinjections
    - F addbookmarks
    - F analyzenormalsymbol
    - F analyzesymbol
    - F attachmentid
    - F bitmap
    - F clonefield
    - F copyimage
    - F defaultprofile
    - F definefield
    - F definefieldset
    - F defineviewerlayer
    - F embedfile
    - F embedimage
    - F enabletags
    - F exportformdata
    - F fieldsurrounding
    - F flushmergelayer
    - F getbookmarks
    - F getboxresourcebox
    - F getdefaultfieldvalue
    - F getfieldcategory
    - F getfieldset
    - F getformatoption
    - F getinfo
    - F getpreviewfigure
    - F getrefobj
    - F insertrenderingwindow
    - F jp2
    - F jpg
    - F maptag
    - F mergebookmarks
    - F mergecomments
    - F mergefields
    - F mergereferences
    - F mergerenditions
    - F mergeviewerlayers
    - F newimage
    - F png
    - F prerollreference
    - F presetsymbol
    - F presetsymbollist
    - F processrendering

F reducetags
F registercompactor
F registerdocumentcloseaction
F registerdocumentopenaction
F registeredsymbol
F registerfiguremask
F registerfigurerange
F registerpagecloseaction
F registerpageopenaction
F registerpattern
F registersymbol
F ruleactionbox
F ruleactiondraw
F ruleactionfill
F ruleactionmp
F ruleactionstroke
F saveboxresource
F scanimage
F setattachmentsupport
F setautoprefix
F setencryption
F setfieldcalculationset
F setfigurealternative
F setfigurecolorspace
F setfiguremask
F setformat
F setformsmethod
F setpagetransition
F settaggingsupport
F setupviewerlayer
F setxmpfile
F startlayer
F startunicodetoactualtext
F startunicodetoactualtextdirect
F stoplayer
F stopunicodetoactualtext
F stopunicodetoactualtextdirect
F supportedformats
F unicodetoactualtext
F useviewerlayer
F validcompactor
F validfield
F validfieldcategory
F validfieldset
F vfstartcolor
F vfstopcolor
F wrapimage
S name
T nodeinjections
F addtags
F annotation
F attachfile
F changestackedlayer
F cmykcolor
F comment
F destination
F effect
F getboxresourcedimensions
F graycolor
F injectbitmap
F insertu3d
F knockout
F negative
F overprint
F positive
F reference
F rgbcolor
F spotcolor
F startalternate
F startlayer
F startstackedlayer
F stopalternate
F stoplayer
F stopstackedlayer
F transparency
F typesetfield
F useboxresource
T registrations
F cmykindexcolor
F cmykspotcolor
F getspotcolorreference
F grayindexcolor
F grayspotcolor
F rgbindexcolor
F rgbspotcolor
F spotcolorname
F transparency
T tables

registrations	table	**writeout**	function
setencryption	function		
tables	table		

basexx (extra lua)

decode16	function	**encode16**	function
decode64	function	**encode64**	function
decode85	function	**encode85**	function
decodeLZW	function	**encodeLZW**	function
decodeRL	function	**encodeRL**	function

bit32 (context)

arshift	function	**extract**	function
band	function	**lrotate**	function
bnot	function	**lshift**	function
bor	function	**replace**	function
btest	function	**rrotate**	function
bxor	function	**rshift**	function

blobs (context)

append	function
dimensions	function
dispose	function
new	function
pack	function
strdp	function
strhd	function
strht	function
strwd	function
tohpack	function
tonodes	function
tovpack	function
write	function

boolean (context)

tocontext function
tonumber function

buffers (context)

append	function
assign	function
collectcontent	function
empty	function
erase	function
exists	function
get	function
getcontent	function
getctxlua	function
getlines	function
getmkvi	function
gettex	function
loadcontent	function
parallel	table
F define	
F hassomecontent	
F next	
F place	
F reset	
F save	
prepend	function
raw	function
run	function
samplefile	function
splitsamplefile	function
undent	function

builders (context)

checkparcontext function
checksidefloat function
hpack table
F qualityactions
F report
F show
kernel table
F collapsing
F hyphenating
F kerning
F ligaturing
F mlisttohlist
local_box_filter function
paragraphs table
T constructors
N attribute
F define
F disable
F enable
F handler
T methods
F basic
F default
F hpack
F oneline
D names
D numbers
F set
F start
F stop
T expansion
F trace
F keeptogether
F registertogether
T solutions
T splitters
F define
F optimize
T registered
F reset
F set
F split
F start
F stop
F tag
profiling table
F add
F addrivers
F boxlinehandler
F distance
F fixedprofile
F get
F getrivers
F list
F pagehandler
F pagelinehandler
F profilebox
F profilelines
F set
F setlines
registerpaspasscallback function
vpack table
F qualityactions
F report
vspacing table
D categories
F checkstrutdepth
F collapsevbox
T data
D map
D skip
D snapmethods
F definesnapmethod
B fixed
F flushcontributions
F forcepageflush
F forcestrutdepth
F getnofpreviouslines
F inject
F injectdisable
F injectpenalty
F injectskip

F interceptsamepagecontent

F pagehandler

F pageoverflow

F popatsame

F pushatsame

F setmap

F setskip

F snapbox

F synchronizepage

F tocategories

F tocategory

F vboxhandler

caches (context)

base	string	**loadcontent**	function
collapsecontent	function	**loaddata**	function
configfiles	function	**more**	string
contentstate	function	**savecontent**	function
defaults	table	**savedata**	function
getfirstreadablefile	function	**setfirstwritablefile**	function
getreadablepaths	function	**setluanames**	function
getwritablepath	function	**treehash**	function
hashed	function	**usedpaths**	function
is_writable	function		

callback (basic tex)

find	function
known	function
list	function
register	overloaded function
usage	function

callbacks (context)

freeze function
functions table
- F start_page_number
- F stop_page_number

known function
list data table
pop function
push function
register function
report function
supported table
- B alignment_filter
- B append_line_filter
- B append_to_vlist_filter
- B begin_paragraph
- B buildpage_filter
- B define_font
- B find_format_file
- B find_log_file
- B get_attribute
- B get_math_dictionary
- B get_noad_class
- B glyph_run
- B handle_overload
- B handle_uleader
- B hpack_filter
- B hpack_quality
- B hyphenate
- B insert_distance
- B insert_par
- B intercept_lua_error
- B intercept_tex_error
- B italic_correction
- B kerning
- B ligaturing
- B linebreak_filter
- B linebreak_quality
- B local_box_filter
- B make_extensible
- B math_rule
- B missing_character
- B mlist_to_hlist
- B open_data_file
- B packed_vbox_filter
- B paragraph_context
- B paragraph_pass
- B post_linebreak_filter
- B pre_dump
- B pre_linebreak_filter
- B pre_output_filter
- B process_character
- B process_jobname
- B quality_font
- B register_extensible
- B show_break
- B show_build
- B show_error_message
- B show_lua_call
- B show_warning_message
- B show_whatsit
- B start_file
- B start_run
- B stop_file
- B stop_run
- B trace_memory
- B vpack_filter
- B vpack_quality
- B wrapup_run

catcodes (context)

names data table

numbers data table

register function

characters (context)

addcharacternames	function
adobename	function
bidi	data table
blockrange	function
blocks	data table
can_have_space	data table
casemap	table
categories	data table
category	function
categorytags	data table
classifiers	table
collapsed	data table
combined	data table
contextname	function
csletters	data table
data	data table
decomposed	data table
description	function
descriptions	data table
detailtags	data table
directions	table
emoji	table
F compact	
F known	
F resolve	
emojized	function
enccodes	data table
entities	data table
fallbacks	data table
filters	data table
fractions	table
fromnumber	function
fschars	data table
fscodes	data table
getrange	function
graphemes	data table
hangul	data table
indicgroups	table
T above_mark	
T after_half	
T after_main	
T after_postscript	
T after_subscript	
T anudatta	
T before_half	
T before_main	
T before_postscript	
T before_subscript	
T below_mark	
T consonant	
T dependent_vowel	
T halant	
T independent_vowel	
T nukta	
T post_mark	
T pre_mark	
T ra	
T stress_tone_mark	
T twopart_mark	
T vowel_modifier	
initialize	function
is_character	data table
is_command	data table
is_hyphenator	data table
is_letter	data table
is_mark	data table
is_punctuation	data table
is_spacing	data table
is_symbol	data table
lcchars	data table
lccode	function
lccodes	data table
lettered	function
lhash	data table
linebreaks	data table
lower	function
mathlists	data table
mirrors	table
otfscripts	data table
ranges	data table
registerentities	function
remap_hangul_syllabe	function

scripthash	table
setactivecatcodes	function
setlettercatcodes	function
setothercatcodes	function
shape	function
shaped	function
shash	data table
shchars	data table
shcodes	data table
showstring	function
specialchars	data table
specials	data table
splits	table
T char	
T compat	
subscripts	table
superscripts	table
synonyms	data table
tex	table
D accentmapping	
D commandmapping	
F defineaccents	
F safechar	
F strtextoutf	
F strtoutfpattern	
F toutf	
F toutfpattern	
textclasses	table
tonumber	function
ucchars	data table
uccode	function
uccodes	data table
uhash	data table
unicodechar	function
upper	function
verbosemarks	data table

chemistry (context)

component	function	**method**	string
define	function	**molecule**	function
definitions	data table	**moleculeparser**	userdata
format	string	**start**	function
inlinechemical	function	**stop**	function
instance	string	**undefine**	function

containers (context)

cleanname	function
content	function
define	function
is_usable	function
is_valid	function
read	function
usecache	boolean
write	function

converters (context)

A	function
A:es	function
A:ru	function
A:sl	function
AK	function
Alphabetic	function
Alphabetic:es	function
Alphabetic:ru	function
Alphabetic:sl	function
Character	function
Characters	function
I	function
KA	function
KR	function
R	function
RK	function
Romannumerals	function
Russiannumerals	function
Sloveniannumerals	function
Spanishnumerals	function
Word	function
Words	function
a	function
a:es	function
a:ru	function
a:sl	function
abjadnodotnumerals	function
abjadnumerals	function
alphabetic	function
alphabetic:es	function
alphabetic:ru	function
alphabetic:sl	function
character	function
characters	function
chineseallnumerals	function
chinesecapnumerals	function
chinesedatenumerals	function
chinesenumerals	function
chr	function
chrs	function
cn	function
cn-a	function
cn-c	function
cn-d	function
convert	function
decimals	function
define	function
gregorian_to_hebrew	function
gregorian_to_jalali	function
hebrewnumerals	function
helpers	table
hexstringtonumber	function
i	function
isleapyear	function
jalali_to_gregorian	function
lchexnumber	function
lchexnumbers	function
leapyear	function
lower	function
max	function
maxchrs	function
nicenumber	function
nofdays	function
number	function
numbers	function
octnumber	function
octstringtonumber	function
ordinal	function
r	function
romannumerals	function
russiannumerals	function
sequences	data table
settime	function
sloveniannumerals	function
spaceddigits	function
spanishnumerals	function
textime	function
toabjad	function
tochinese	function
tohebrew	function
toroman	function
totime	function

uchexnumber	function	F translate	
uchexnumbers	function	**weekday**	function
upper	function	**word**	function
verbose	table	**words**	function

D data

coroutine (basic lua)

close	function	**running**	function
create	function	**status**	function
isyieldable	function	**wrap**	function
resume	function	**yield**	function

ctxrunner (context)

load	function	**prepfiles**	data table
preparedfile	function	**resolve**	function

debug (basic lua)

getinfo function
sethook function
traceback function

dir (context)

collectpattern	function
current	function
expandlink	function
expandname	function
found	function
glob	function
globdirs	function
globfiles	function
globpattern	function
ls	function
makedirs	function
mkdirs	function
pop	function
push	function

directives (context)

data	data table
default	function
disable	function
done	table
enable	function
list	function
name	string
register	function
report	function
reset	function
show	function
value	function
values	function

document (context)

arguments	data table
files	table
functions	table
getargument	function
getfilename	function
options	table

T commandline

D environments

D modes

D modules

T ctxfile

D environments

D modes

D modules

pdf_add_cidsets	function
pdf_identify_content	function
pdf_strip_content	function
pdf_strip_page	function
setargument	function
setcommandline	function
setdefaultargument	function
setenvironments	function
setfilename	function
setfilenames	function
setmodes	function
setmodules	function
variables	data table

documentdata (context)

variable function

drivers (context)

applyanchor function
applyorientation function
convert function
converters table
  F analyze
  F engine
  F lmtx
  F resyncbox
enable function
gethpos function
getpos function
getrpos function
getvpos function
helpers table
install function
instances table
  T empty
    T actions
      F convert
      F prepare
    T flushers
      F popleaderlevel
      F pushleaderlevel
      T whatsit
    S name
  T js
    T actions
      F cleanup
      F convert
      F finalize
      F initialize
      F outputfilename
      F prepare
      F wrapup
    T flushers
      F character
      F popleaderlevel
      F pushleaderlevel
      F rule
      F simplerule
      F specialrule
      F updatefontstate
      T whatsit
    S name
  T json
    T actions
      F cleanup
      F convert
      F finalize
      F initialize
      F outputfilename
      F prepare
      F wrapup
    T flushers
      F character
      F popleaderlevel
      F pushleaderlevel
      F rule
      F setstate
      F simplerule
      F specialrule
      F updatefontstate
      T whatsit
    S name
  T lua
    T actions
      F cleanup
      F convert
      F finalize
      F initialize
      F outputfilename
      F prepare
      F wrapup
    T flushers
      F character
      F popleaderlevel
      F pushleaderlevel
      F rule
      F simplerule
      F specialrule
      F updatefontstate
      T whatsit

```
  S name
T mpo
  T actions
    F finalize
    F initialize
  T flushers
    F character
    F popleaderlevel
    F pushleaderlevel
    F rule
    F simplerule
    F specialrule
    T whatsit
  S name
T mps
  T actions
    F cleanup
    F convert
    F finalize
    F initialize
    F outputfilename
    F prepare
    F wrapup
  T flushers
    F character
    F popleaderlevel
    F pushleaderlevel
    F rule
    F simplerule
    F specialrule
    F updatefontstate
    T whatsit
  S name
T none
  T actions
  T flushers
    F popleaderlevel
    F pushleaderlevel
    T whatsit
  S name
T pdf
  T actions
    F cleanup
    F convert
    F finalize
    F initialize
    F outputfilename
    F prepare
    F wrapup
  T flushers
    F character
    F fontchar
    F group
    F image
    F literal
    F popleaderlevel
    F poporientation
    F pushleaderlevel
    F pushorientation
    F restore
    F rule
    F save
    F setmatrix
    F setstate
    F simplerule
    F space
    F specialrule
    F startclipping
    F startmatrix
    F startmirroring
    F startrotation
    F startscaling
    F stopclipping
    F stopmatrix
    F stopmirroring
    F stoprotation
    F stopscaling
    F updatefontstate
    T whatsit
  S name
```

lmtxversion	number
outputfilename	function
report	function

environment (context)

argument	function	**luafilechunk**	function
arguments	data table	**luatexengine**	string
basicengines	data table	**luatexfunctionality**	number
currentrun	number	**luatexverbose**	string
definefilefallback	function	**luatexversion**	number
definefilesynonym	function	**maxnofruns**	number
engineflags	data table	**oldhome**	string
filename	string	**originalarguments**	data table
filenames	table	**outputfilename**	string
files	table	**ownbin**	string
filesynonyms	data table	**ownmain**	string
forcedruns	number	**ownname**	string
getargument	function	**ownpath**	string
homedir	string	**prerollrun**	boolean
initex	boolean	**rawarguments**	data table
initexmode	boolean	**reconstructcommandline**	function
initializearguments	function	**relativepath**	function
initializefilenames	function	**setargument**	function
inputfilebarename	string	**splitarguments**	function
inputfilename	string	**suffix**	string
inputfilesuffix	string	**texfile**	function
jobfilefullname	string	**texmfos**	string
jobfilename	string	**texos**	string
jobfilesuffix	string	**texroot**	string
kindofrun	number	**truefilename**	function
lmtxmode	number	**useluamodule**	function
loadexamodes	function	**usemodules**	function
loadluafile	function	**validengines**	data table
luafile	function	**version**	string

experiments (context)

data	data table	**register**	function
default	function	**report**	function
disable	function	**reset**	function
enable	function	**show**	function
list	function	**value**	function
name	string	**values**	function

figures (context)

applyratio	function
badname	function
bases	data table
bitmapimage	function
boxnumber	number
cachepaths	data table
check	function
checkers	data table
conversions	data table
converters	data table
current	function
defaultdepth	number
defaultheight	number
defaultsearch	boolean
defaultwidth	number
done	function
dummy	function
existers	data table
exists	function
formats	data table
found	data table
get	function
getinfo	function
getpdfinfo	function
getrealpage	function
guess	function
guessfromstring	function
hash	function
identifiers	data table
identify	function
images	table

- F check
- F checksize
- F copy
- F create
- F embed
- F embedfromstore
- F fetchdata
- T keys
- F print
- F resetdata
- F resetstore
- F scan
- F serialize
- T sizes
  - B art
  - B bleed
  - B crop
  - B media
  - B none
  - B trim
- F storedata
- F totable
- T types
  - N jbig2
  - N jp2
  - N jpg
  - N none
  - N pdf
  - N png
  - N stream
- F wrap

include	function
includers	data table
initialize	function
loaded	data table
localpaths	data table
magics	data table
mergegoodies	function
nofmissing	number
nofprocessed	number
order	data table
paths	data table
patterns	data table
pop	function
preferquality	boolean
programs	data table
push	function
registerpattern	function
registerresource	function
registersuffix	function
remappers	data table

resources	data table	**setorder**	function
scale	function	**setpaths**	function
set	function	**suffixes**	data table
setconversion	function	**used**	data table
setlookups	function		

file (basic lua)

addsuffix	function	**needsupdating**	function
basename	function	**pathpart**	function
checksum	function	**readdata**	function
collapsepath	function	**removesuffix**	function
copy	function	**replacesuffix**	function
dirname	function	**reslash**	function
expandname	function	**robustname**	function
extname	function	**savechecksum**	function
is_qualified_path	function	**savedata**	function
is_readable	function	**size**	function
is_rootbased_path	function	**splitbase**	function
is_writable	function	**splitname**	function
isreadable	function	**splitpath**	function
iswritable	function	**strip**	function
join	function	**suffix**	function
joinpath	function	**suffixes**	function
loadchecksum	function	**suffixesonly**	function
nameonly	function	**suffixonly**	function
nametotable	function	**syncmtimes**	function
needs_updating	function	**withinbase**	function

fio (extra lua)

getposition	function	**readinteger1le**	function
read2dot14	function	**readinteger2**	function
readbytes	function	**readinteger2le**	function
readbytetable	function	**readinteger3**	function
readcardinal1	function	**readinteger3le**	function
readcardinal1le	function	**readinteger4**	function
readcardinal2	function	**readinteger4le**	function
readcardinal2le	function	**readintegertable**	function
readcardinal3	function	**setposition**	function
readcardinal3le	function	**skipposition**	function
readcardinal4	function	**writecardinal1**	function
readcardinal4le	function	**writecardinal1le**	function
readcardinaltable	function	**writecardinal2**	function
readcline	function	**writecardinal2le**	function
readcstring	function	**writecardinal3**	function
readfixed2	function	**writecardinal3le**	function
readfixed4	function	**writecardinal4**	function
readinteger1	function	**writecardinal4le**	function

floats (context)

analysemethod	function	**nofstacked**	function
checkedpagefloat	function	**pop**	function
collect	function	**push**	function
consult	function	**resave**	function
flush	function	**save**	function
getvariable	function	**stacked**	function
hasstacked	function		

font (basic tex)

addcharacters	function	**getmathspec**	function
addquality	function	**id**	function
current	function	**max**	function
define	function	**nextid**	function
getfont	function	**setfont**	function
getfontdimen	function	**setfontdimen**	function
getfontspec	function	**xscaled**	function
getmathindex	function	**yscaled**	function

fonts (context)

analyzers table
- D features
- D initializers
- D methods
- F setstate
- D states
- B useunicodemarks

callbacks table
- F callback
- F devirtualize

checkers table
- F getmissing
- D mapping
- F missing
- F placeholder

cid table
- F getmap
- F loadfile

collections table
- F checkenabled
- F clonevector
- F define
- T definitions
- F direct
- F found
- F prepare
- F process
- F register
- F registermain
- F report
- F reset
- T rscales
- T vectors

constructors table
- F addcharacters
- F addcoreunicodes
- F aftercopyingcharacters
- F applyfinalizers
- F applymanipulators
- F assignmathparameters
- F beforecopyingcharacters
- F beforepassingfonttotex
- D cache
- F calculatescale
- F checkedfeatures
- F checkedfilename
- F collectprocessors
- F currentfonthasfeature
- D designsizes
- F enhanceparameters
- T enhancers
  - T afm
    - F apply
    - F patch
    - T patches
      - F register
      - F report
    - F register
    - F report
  - T otf
    - F addfeature
    - F apply
    - F enhance
    - F patch
    - T patches
      - F register
      - F report
    - F register
    - F report
  - T tfm
    - F apply
    - F patch
    - T patches
      - F register
      - F report
    - F register
    - F report
- N factor
- T features
  - D afm
  - D otf
  - D tfm

```
F finalize
F getfeatureaction
F getmathparameter
F getprivate
T handlers
  T afm
    B autoprefixed
    D cache
    D features
    F getkern
    T helpdata
      D bothkerned
      D leftkerned
      D ligatures
      D rightkerned
      D texligatures
    F load
    T readers
      F getinfo
      F loadfont
      F loadshapes
    F setfeatures
    B syncspace
    N version
  T otf
    F addfeature
    F basemodeinitializer
    D cache
    F collectlookups
    T coverup
      T actions
        F alternate
        F chainposition
        F chainsubstitution
        F kern
        F ligature
        F multiple
        F pair
        F single
        F substitution
      F register
      S stepkey
    F dataset
    F datasetpositionprocessor
    S defaultbasealternate
    S defaultnodealternate
    D features
    F featuresinitializer
    F featuresprocessor
    F getactualtext
    F getalternate
    F getgsub
    F getkern
    F getmultiple
    F getstreamhash
    F getsubstitution
    T handlers
      F ctx_camelcasing
      F devanagari_remove_joiners
      F devanagari_reorder_matras
      F devanagari_reorder_pre_base_reordering_consonants
      F devanagari_reorder_reph
      F gpos_context
      F gpos_contextchain
      F gpos_cursive
      F gpos_mark2base
      F gpos_mark2ligature
      F gpos_mark2mark
      F gpos_pair
      F gpos_single
      F gsub_alternate
      F gsub_context
      F gsub_contextchain
      F gsub_ligature
      F gsub_multiple
      F gsub_reversecontextchain
      F gsub_single
      F trigger_space_kerns
    T helpers
      F blockligatures
      F pardirstate
      F txtdirstate
    F load
    F loadoutlinedata
    F loadstreamdata
    F modeinitializer
    D mpscache
    D pdfcache
```

F pluginitializer
F pluginprocessor
T plugins
  T text
D pngcache
B pngenabled
T readers
  F addunicodetable
  F avar
  F cbdt
  F cblc
  F cff
  F cff2
  T cffencoder
  F checkhash
  F cmap
  F colr
  F compact
  F condense
  F cpal
  F dsig
  F ebdt
  F eblc
  F ebsc
  F expand
  F extend
  F fvar
  F gdef
  F getcomponents
  F getinfo
  F glyf
  F gpos
  T gposhandlers
    F chainedcontext
    F context
    F cursive
    F extension
    F marktobase
    F marktoligature
    F marktomark
    F pair
    F single
  F gsub
  T gsubhandlers
    F alternate
    F chainedcontext
    F context
    F extension
    F ligature
    F multiple
    F reversechainedcontextsingle
    F single
  F gvar
  F head
  T helpers
    F axistofactors
    F cleanname
    F getaxisscale
    F getfactors
    F getscales
    F gotodatatable
    F normalizedaxis
    T normalizedaxishash
    F readvariationdata
    F reportskippedtable
    F setvariabledata
  F hhea
  F hmtx
  F hvar
  F kern
  F loadfont
  F loadshapes
  F loca
  F math
  F maxp
  F mvar
  F name
  F os/2
  F pack
  F parsecharstrings
  F post
  F potracetocff
  F registerextender
  F rehash
  F sbix
  F stat
  T streamreader
    F close

F getposition
F getsize
F look
F open
F read2dot14
F readbyte
F readbytes
F readbytetable
F readcardinal
F readcardinal1
F readcardinal1le
F readcardinal2
F readcardinal2le
F readcardinal3
F readcardinal3le
F readcardinal4
F readcardinal4le
F readcardinaltable
F readchar
F readcline
F readcstring
F readfixed2
F readfixed4
F readinteger
F readinteger1
F readinteger1le
F readinteger2
F readinteger2le
F readinteger3
F readinteger3le
F readinteger4
F readinteger4le
F readintegertable
F readsignedbyte
F readstring
F readtag
F setposition
F size
F skip
F skiplong
F skipposition
F skipshort
F writebyte
F writecardinal1
F writecardinal1le
F writecardinal2
F writecardinal2le
F writecardinal3
F writecardinal3le
F writecardinal4
F writecardinal4le
F writeinteger1
F writeinteger1le
F writeinteger2
F writeinteger2le
F writeinteger3
F writeinteger3le
F writeinteger4
F writeinteger4le
F writestring
F stripredundant
F svg
N tableversion
F unifymissing
F unpack
F vhea
F vmtx
F vorg
F vvar
F woff2otf
F registerpalette
F registerplugin
F reorderlookups
F scriptandlanguage
F setdynamics
F setfeatures
T statistics
D usedfeatures
D svgcache
B svgenabled
T tables
D baselines
D features
D featuretypes
D languages
D scripts
F texthandler
T vectors

```
    T arab
      T gpos
        N curs
        N kern
        N mark
        N mkmk
      T gsub
        N calt
        N ccmp
        N cswh
        N dlig
        N fina
        N init
        N isol
        N liga
        N medi
        N mset
        N rclt
        N rlig
  N version
T pfb
  F filterencoding
  F filtermetadata
  F loadvector
  N version
T tfm
  D cache
  D features
  N maxnestingdepth
  N maxnestingsize
  T readers
    F loadpk
    F loadtfm
    F loadtfmvf
    F loadvf
    F pktopdf
    F potracedtopdf
    F showpk
  F reencode
  F setfeatures
  N version
T vf
  T combinations
  F combine
    T combiner
      D commands
    D helpers
    D math
    D predefined
    N version
    D whatever
  F hashfeatures
  F hashinstance
  T hashmethods
    F normal
    F virtual
  F initializefeatures
  D loadedfonts
  N noffontsloaded
  N nofsharedfonts
  N nofsharedhashes
  N nofsharedvectors
  N privateoffset
  D privateslots
  F readanddefine
  F registerfeature
  F resetfeature
  B resolvevirtualtoo
  F scale
  F scaled
  F setfactor
  F setfeature
  F setmathparameter
  F setname
  B sharefonts
  F trytosharefont
  N version
```

current function

currentid function

definers table

```
  F addlookup
  F analyze
  F applypostprocessors
  F checkedmode
  F checkedscript
  S defaultlookup
  F define
  F getspecification
```

```
  F internal
  F loadfont
  F makespecification
  T methods
    F install
    D variants
  F read
  F register
  F registered
  F registerfallbackfont
  F registersplit
  F resetnullfont
  F resolve
  T resolvers
    F file
    F name
    F spec
definetypeface    function
dropins           table
  F clone
  F getshape
  F getshapes
  F nextid
  F provide
  F registerglyph
  F registerglyphs
  F swap
  F swapone
effects           table
encodings         table
  T agl
    D ctxcodes
    D extras
    D synonyms
    D unicodes
  D cache
  F is_known
  D known
  F load
  F make_unicode_vector
  D math
  N version
enhanceid         function
expansions        data table
formats           data table
getalternate      function
getindices        function
getkern           function
getmultiple       function
getsubstitution   function
getters           table
  T alternate
    F opentype
  T kern
    F opentype
    F type1
    F type3
  T multiple
    F opentype
  T substitution
    F opentype
goodies           table
  D colorschemes
  T data
    T common-math.lfg
      S author
      S comment
      S copyright
      B initialized
      T mathematics
        T tweaks
          T presets
            F addmathkerns
            F braille
            F eulertocalligraphic
            F fallbacks
            F moderntocalligraphic
            F moveintegrals
            F moveitalics
            F pagellaradical
            F rsfstoscript
            F rsfsuprighttoscript
            F scripttocalligraphic
            F smalldigits
            F wipeanchors
            F wipeitalics
            F xitsarabic
      S name
```

```
    S version
  T dejavu-math
    S author
    S comment
    S copyright
    B initialized
    T mathematics
      T alternates
        T dotless
          S comment
          S feature
          N value
      T bigslots
      T parameters
        N AccentBaseDepth
        N AccentTopShiftUp
        N DelimiterDisplayPercent
        N DelimiterDisplayShortfall
        N DelimiterPercent
        N DelimiterShortfall
        N DisplayOperatorMinHeight
        N FlattenedAccentTopShiftUp
        N FractionRuleThickness
        S PrimeBaselineDropMax
        S PrimeShiftUp
        S PrimeShiftUpCramped
        N SubscriptShiftDown
        S SubscriptShiftDownWithSuperscript
        N SuperscriptShiftUp
      T tweaks
        T aftercopying
    S name
    S version
  T dejavu-text
    S author
    S comment
    S copyright
    T extensions
    B initialized
    S name
    S version
D designsizes
D filenames
F getcompositions
    T list
    F load
    F prepare_features
    F register
    F registerpostprocessor
    F report
    T suffixes
handlers            table
hashes              table
    D characters
    D classes
    D csnames
    D descriptions
    D dynamics
    D emwidths
    D exheights
    D features
    D identifiers
    D italics
    D lastmathids
    D marks
    D mathparameters
    D modes
    D originals
    D parameters
    T processes
    D properties
    D quads
    D resources
    T setdynamics
    T shapes
    D spaces
    D spacings
    T streams
    D unicodes
    D unislots
    D variants
    D xheights
helpers             table
    F addextraprivate
    F addprivate
    F appendcommands
    F appendcommandtable
    F checkflags
```

F checkmerge
F checksteps
F collectanchors
F combineglyphs
D commands
F composecharacters
F descriptiontoslot
F dimenfactor
F expandglyph
D extraprivates
F fontpatternhassize
F getcoloredglyphs
F getdigitwidth
F getinstancespec
F getparameters
F getprivatecharornode
F getprivatenode
F getprivates
F getprivateslot
F hasprivate
F indextoslot
F isprivate
F mathscriptslots
F name
F nametoslot
F newprivateslot
F prependcommands
F prependcommandtable
F privateslot
F registerglyphbox
F scalecommands
F setboxdirectly
F setboxglyph
F setboxglyphs
F setdigitwidth
F setmetaglyph
F setmetaglyphs
D sharedprivates
F splitfontpattern
F tochar
T vfinjectors
  F startcolor
  F stopcolor
T vfspecials
  T backgrounds
  T outlines

infofont function

iterators table
  F characters
  F descriptions
  F glyphs

loggers table
  F format
  F onetimememessage
  F register
  F reportdefinedfonts
  F reportusedfeatures

mappings table
  F addtounicode
  F fromunicode16
  F getentry
  F loadfile
  F loadline
  F makenameparser
  D overloads
  F reset
  F tounicode
  F tounicode16
  F tounicode16sequence

metapost table
  F baseline
  F boundingbox
  F boxtomp
  F draw
  F fill
  F glyph
  F maxbounds
  F paths
  F widthline
  F zeroline

mode function

names data table

nulldata data table

privateoffsets table
  B keepnames
  N mathbase
  N mathextrabase
  N textbase

N textextrabase

protrusions data table

readers table

F afm

F check_lua

F check_tfm

F lua

F ofm

F opentype

F otf

F pfb

D sequence

F tfm

F ttc

F ttf

F woff

registersanitizer function

reserveid function

select table

D alternatives

F bodyfont

D data

F default

D defaults

F definefallbackfamily

F definefontfamily

F definefontfamilypreset

D extras

F fallback

D fallbacks

F features

F filefallback

F fontdata

F fontfallback

F fontsynonym

F goodies

F mathfallback

D methods

D presets

F registerfallback

F registerfiles

F registerfontalternative

F registerfontfamily

F textfallback

F typeface

F typescript

F userdata

specifiers table

D contextmerged

F contextnumber

D contextnumbers

D contextsetups

F contexttostring

F definecontext

F mergecontext

F presetcontext

F registercontext

F showcontext

F splitcontext

D synonyms

D variants

symbols table

T emoji

F checked

F resolved

F uselibrary

tables table

toeffect function

tracers table

treatments table

F applyfixes

D data

S filename

F ignoredfile

F report

F trace

typefaces table

functions (context)

dummy function

graphics (context)

backgrounds table
- F fetchmultipar
- F pair
- F path
- F point

bitmaps table
- F flush
- F new
- F placeholder
- F tocontext

identifiers table
- F bitmap
- F jp2
- F jpg
- F png

identify function

registerpdfcompactor function

registerpdffont function

synchronizers table
- F collapse
- F collect
- F extend
- F getbottom
- F getd
- F geth
- F getkind
- F getsize
- F gettask
- F gettop
- F getw
- F getx
- F gety
- F prune

gzip (context)

compress	function	**load**	function
compressed	function	**save**	function
decompress	function	**suffix**	function

images (context)

check function
checksize function
copy function
create function
embed function
embedfromstore function
fetchdata function
keys table
print function
resetdata function
resetstore function
scan function
serialize function
sizes table
- B art
- B bleed
- B crop
- B media
- B none
- B trim

storedata function
totable function
types table
- N jbig2
- N jp2
- N jpg
- N none
- N pdf
- N png
- N stream

wrap function

interactions (context)

attachments data table

fields table

- F clone
- F define
- F defineset
- F insert

general table

- F getidentity
- F getidentityvariable

help table

- F collect
- F reference
- F register

javascripts table

- F addtopreamble
- F code
- D codes
- F flushpreambles
- D functions
- D preambles
- F setpreamble
- F storecode
- F storepreamble
- F usepreamblenow
- F usescripts

linkedlists data table

pages table

- F setpagetransition
- F setupcanvas

references table

- F setclosedocumentaction
- F setclosepageaction
- F setopendocumentaction
- F setopenpageaction

renderings data table

interfaces (context)

arrays	table
cachedsetups	data table
cachesetup	function
cardinals	table
checkedspecification	function
complete	data table
constants	data table
corenamespaces	data table
currentinterface	string
currentresponse	string
definecommand	function
defined	function
doifelsemessage	function
elements	data table
floats	table
formats	data table
getmessage	function
getnamespace	function
getsetupstring	function
hashedsetups	data table
implement	function
integers	table
interfacedcommand	function
interfaces	table
knownscanner	function
message	function
nameofscanner	function
namesofscanners	table
namespaces	table
F define	
F list	
numbers	table
registernamespace	function
registerscanner	function
scanners	data table
setconstant	function
setcount	function
setdimen	function
setelement	function
setformat	function
setmacro	function
setmessage	function
setmessages	function
settranslation	function
setupstrings	data table
setvariable	function
showmessage	function
storedscanners	data table
tolist	function
translations	data table
usednamespaces	data table
variables	data table
writestatus	function

io (basic lua)

ask	function
bytes	function
characters	function
close	function
copydata	function
exists	function
fileseparator	string
flush	function
gobble	function
input	overloaded function
lines	overloaded function
loadchunk	function
loaddata	function
loadlines	function
newreader	function
noflines	function
open	overloaded function
output	overloaded function
pathseparator	string
popen	overloaded function
read	function
readall	function
readnumber	function
readstring	function
savedata	function
size	function
stderr	userdata
stdin	userdata
stdout	userdata
tmpfile	function
type	function
write	function

job (context)

comment function
datasets table
- D collected
- F extend
- F getdata
- F setdata
- D tobesaved

disablesave function
embedded table
- D collected
- D tobesaved

fileobjreferences table
- D collected
- D tobesaved

files table
- D collected
- F context
- B forcerun
- F run
- D tobesaved

getrandomseed function
initialize function
linkedlists table
- D collected
- D tobesaved

load function
loadother function
objects table
- D collected
- F get
- F number
- F page
- F save
- F set
- D tobesaved

pack boolean
packversion number
pagestates table
- D collected
- F countervalue
- F extend
- F realpage
- F realpageorder
- F setstate
- D tobesaved

parwrappers table
- D collected
- D tobesaved

positions table
- F b_column
- F b_region
- F c
- D collected
- F column
- F columnofpos
- F copy
- F depth
- F e_column
- F e_region
- F enhance
- F extra
- F get
- F getcolumndata
- F getfree
- F gethpos
- F getpos
- F getreserved
- F getrpos
- F getsync
- F gettobesaved
- F getvpos
- F hangafter
- F hangindent
- F height
- F hsize
- N lastx
- N lasty
- F leftskip
- F lowerleft
- F lowerright
- F markregionbox
- F n

F onsamepage
F overlapping
F p
F page
F paragraph
F parindent
F position
F r
F region
F registerhandlers
F replace
F rightskip
F set
F setall
F setregionbox
F setspec
F settobesaved
D tobesaved
F upperleft
F upperright
F used
F whd
F width
F x
F xy
F y

register function

save function

structure table

D collected
T components
T namestack
T tobesaved
T branches
S name
S type

variables table

D checksums
D collected
F getchecksum
F makechecksum
F restore
F save
F setchecksum
D tobesaved

version number

language (basic tex)

clean	function	**id**	function
clearhyphenation	function	**new**	function
clearpatterns	function	**patterns**	function
current	function	**postexhyphenchar**	function
gethjcode	function	**posthyphenchar**	function
has_language	function	**preexhyphenchar**	function
hyphenate	function	**prehyphenchar**	function
hyphenation	function	**sethjcode**	function
hyphenationmin	function	**setwordhandler**	function

languages (context)

addgoodiesdata function
addgoodiesfile function
appliedoptions table
associate function
associated data table
association function
counters data table
current function
data data table
decimals data table
define function
explicithyphen function
flatten function
frequencies table
F averagecharwidth
F getdata
getdata function
getnumber function
goodiefiles function
hjcounts table
hyphenate function
hyphenators table
T dictionaries
D featuresets
F handler
T methods
F expanded
B none
F original
F tex
F traditional
F popmethod
F pushmethod
F setmethod
F setup
F showhyphenationtrace
T traditional
F definefeatures
F gettrace
F hyphenate
F injecthyphens
F installmethod
F lasttrace
F loadpatterns
F registerexception
F registerpattern
F setfeatures
installed function
installhandler function
labels table
F define
F split
loadable function
loadwords function
logger table
F report
nofflattened function
numbers data table
postexhyphenchar function
posthyphenchar function
preexhyphenchar function
prehyphenchar function
registered data table
registeredactions table
replacements table
F add
F addlist
F handler
F set
serializediscretionary function
setexceptions function
setgoodieshandler function
setoptions function
setpatterns function
setsynonym function
setwordhandler function
showdiscretionaries function
strippedgoodiewords function
tolang function
transliteration table
F exception
F handler

F register
F set
F use

unload function

urls table

D characters
N lefthyphenmin
D mapping
B packslashes
N righthyphenmin
F setcharacters

version number

visualizediscretionaries function

words table

F check
T data
F disable
F enable
T enablers
B enables
F found
F load
T methods
N threshold

layouts (context)

getcanvas	function
getpagedimensions	function
setupcanvas	function

status table

- F isleftpage
- F leftorrightpageaction

lfs (extra lua)

attributes	overloaded function	**link**	overloaded function
chdir	overloaded function	**mkdir**	overloaded function
currentdir	function	**mkdirs**	function
dir	overloaded function	**modification**	function
isdir	overloaded function	**readlink**	function
isfile	overloaded function	**rmdir**	overloaded function
isfound	function	**setexecutable**	function
isreadabledir	function	**symlink**	function
isreadablefile	function	**symlinktarget**	function
iswriteabledir	function	**touch**	overloaded function
iswriteablefile	function	**walkdir**	function

libraries (context)

basiclua	data table	**extratex**	data table
basictex	data table	**functions**	data table
builtin	data table	**globals**	data table
extralua	data table	**optional**	data table

lmx (context)

color function
convert function
convertfile function
convertstring function
einject function
eprint function
escape function
finalized function
finject function
fprint function
get function
getvariables function
htmfile function
html table
- F a
- F td
- F th

include function
inherited table
- S color-background-four
- S color-background-one
- S color-background-three
- S color-background-two
- S title

initialize function
inject function
lmxfile function
loadedfile function
make function
new function
overloaderror function
popupfile function
print function
pv function
report function
reset function
result function
set function
show function
showerror function
stripcss function
tv function
type function
urlescape function
variable function
variables table
- S color-background-blue
- S color-background-body
- S color-background-green
- S color-background-main
- S color-background-purple
- S color-background-yellow
- S title-default

logs (context)

application	function
categories	function
direct	function
disable	function
enable	function
finalactions	function
flush	function
loggingerrors	function
messenger	function
new	function
newline	function
poplogfile	function
poptarget	function
private	table
F enablepagetiming	
F getpagetiming	
pushlogfile	function
pushtarget	function
registerfinalactions	function
report	function
reporter	function
resetlogfile	function
setformats	function
setformatters	function
setlogfile	function
setmessenger	function
setprocessor	function
settarget	function
settimedlog	function
settranslations	function
show	function
start_page_number	function
start_run	function
starterrorlogging	function
startfilelogging	function
status	function
stop_page_number	function
stoperrorlogging	function
stopfilelogging	function
subdirect	function
subreport	function
system	function
texerrormessage	function
writer	function

lpdf (context)

action	function
adddocumentcolorspace	function
adddocumentextgstate	function
adddocumentpattern	function
adddocumentshade	function
addtocatalog	function
addtoinfo	function
addtonames	function
addtopageattributes	function
addtopageresources	function
addtopagesattributes	function
addxmpinfo	function
annotationspecification	function
array	function
attachmentsymbols	function
boolean	function
border	function
checkedkey	function
checkedvalue	function
circularshade	function
close	function
cmykcode	function
collectedresources	function
color	function
colorspace	function
colorspaceconstants	data table
colorspec	function
colorvalue	function
colorvalues	function
commentsymbols	function
compresslevel	function
constant	function
copyarray	function
copyboolean	function
copyconstant	function
copydictionary	function
copyfloat	function
copyinteger	function
copynumber	function
copystring	function
copyunicode	function
deferredobject	function
delayedobject	function
destinationspecification	function
dictionary	function
docstring	function
epdf	table

```
  F allcontent
  F analyzefonts
  F contentplugin
  F contenttostring
  F contenttotext
  D encryptioncodes
  F expanded
  F expandwidths
  F fontplugin
  F getpagecontent
  F getstructure
  T helpers
    F tounicodetable
  T image
    F close
    F copy
    F new
    F open
    F query
  F load
  F mergewidths
  D objectcodes
  F objectrange
  F objecttype
  F pageplugin
  F parsecontent
  F producer
  T resolvers
    F destinations
    F embeddedfiles
    F javascripts
    F layers
    F pages
    F structure
    F widgets
```

F stripcontent
F unload
F verboseobject

escaped function

finalizedocument function

finalizepage function

finalizesignature function

finishtransparencycode function

flags table

T annotations
N AutoView
N Hidden
N Invisible
N Locked
N LockedContents
N NoRotate
N NoView
N NoZoom
N Print
N Printable
N ReadOnly
N ToggleNoView
N auto
N hidden
N printable
T widgets
N Comb
N CommitOnSelChange
N DoNotScroll
N DoNotSpellCheck
N Edit
N FileSelect
N MultiLine
N NoExport
N NoToggleToOff
N Password
N PopUp
N PushButton
N Radio
N RadiosInUnison
N ReadOnly
N Required
N RichText
N Sort
N file
N fixed
N nocheck
N protected
N readonly
N required
N sorted
N unavailable

flushfonts function

flushgroup function

flushobject function

flushstreamfileobject function

flushstreamobject function

fonts table

F color_direct
F color_indirect

formats data table

frombytes function

fromeight function

frompdfdoc function

fromsixteen function

getcatalog function

getcompression function

getfontname function

getfontobjectnumber function

getinfo function

getmatrix function

getnofpages function

getpageproperties function

getpagereference function

getpagetags function

getpos function

gettrailerid function

getversion function

getxformname function

graycode function

h_hex_2 table

S
S
S
S
S
S (
S)

S \

h_hex_4 table

hasmatrix function

immediateobject function

includeimage function

initialize function

injectors table

F bitmap

F jp2

F jpg

F png

injectxmpinfo function

insertxmpinfo function

lastreferredpage function

layerreference function

limited function

linearshade function

literal function

majorversion function

makenametree function

minorversion function

nofpages function

null function

number function

object function

objectcompresslevel function

pagereference function

patternstream function

pdfcolor function

pdftimestamp function

preparesignature function

print function

procset function

protectresources boolean

rectangle function

reference function

registerannotation function

registerdestination function

registerdocumentfinalizer function

registerfontmethod function

registerinitializer function

registerpagefinalizer function

registerpattern function

registersignature function

replacexmpinfo function

reserveobject function

rgbcode function

rotationcm function

setcompression function

setencryption function

setincludecidset function

setpageorder function

setpagestate function

setpagetag function

setversion function

sharedobject function

shareobject function

shareobjectreference function

spotcode function

stream function

string function

tocidset function

toeight function

topdfdoc function

tosixteen function

transparency function

transparencycode function

transparencygroups table

B none

transparencyvalue function

unicode function

usedcharacters table

usedfontnames table

N dejavusansmonoboldttf-1-0-0

usedfontobjects table

N dejavusansmonoboldttf-1-0-0

usedfontstreams data table

usedindices table

T dejavusansmonoboldttf-1-0-0

verbose function

lpeg (basic lua)

B	function	**firstofsplit**	function
C	function	**frontstripper**	function
Carg	function	**instringchecker**	function
Cb	function	**is_lpeg**	function
Cc	function	**keeper**	function
Cf	function	**locale**	function
Cg	function	**match**	function
Cmt	function	**oneof**	function
Cp	function	**patterns**	data table
Cs	function	**pcode**	function
Ct	function	**print**	function
P	function	**ptree**	function
R	function	**replacer**	function
S	function	**secondofsplit**	function
UP	function	**setmaxstack**	function
UR	function	**setutfcasers**	function
US	function	**split**	function
V	function	**splitat**	function
afterprefix	function	**splitter**	function
anywhere	function	**stripper**	function
append	function	**times**	function
balancer	function	**tsplitat**	function
beforesuffix	function	**tsplitter**	function
checkedsplit	function	**type**	function
containsws	function	**utfR**	function
counter	function	**utfchartabletopattern**	function
endstripper	function	**utfreplacer**	function
finder	function	**version**	string

lua (basic tex)

bytecode	table	**getstartupfile**	function
bytedata	data table	**getversion**	function
bytedone	table	**lastbytecode**	number
callbytecode	function	**messages**	data table
finalizeinitex	function	**newindex**	function
firstbytecode	number	**newtable**	function
get_functions_table	function	**numbers**	data table
getbytecode	function	**registercode**	function
getcurrenttime	function	**registeredcodes**	function
getdebuginfo	function	**registerinitexfinalizer**	function
getexitcode	function	**registername**	function
getfunctionstable	overloaded function	**setbytecode**	function
getpreciseseconds	function	**setexitcode**	function
getpreciseticks	function	**startupfile**	string
getruntime	function	**version**	string
getstacktop	function		

luac (extra lua)

inspect	function
print	function

luatex (context)

abort function
cleanup function
cleanuptempfiles function
currentfile function
currentlevel function
currenttotal function
poprunstate function
pushrunstate function
registerdumpactions function
registerpageactions function
registerstartactions function
registerstopactions function
registertempfile function
starttime number
synctex table

F blockfilename
F collect
F disable
F enable
F finish
F pause
F popline
F pushline
F registerdisabler
F registerenabler
F resetfilename
F resume
F setfilename
F setup
F start
F stop
F wrapup

wrapup function

lxml (context)

addindex	function
all	function
applyfunction	function
applyselectors	function
att	function
attribute	function
badinclusions	function
chainattribute	function
checkindex	function
collected	function
command	function
concat	function
concatrange	function
content	function
context	function
convert	function
count	function
delete	function
depth	function
direct	function
directives	table
F after	
F before	
F handle	
F load	
F setup	
displayverbatim	function
doif	function
doifelse	function
doifelseempty	function
doifelsetext	function
doifempty	function
doifnot	function
doifnotempty	function
doifnottext	function
doiftext	function
element	function
elements	function
entities	data table
fileline	function
filename	function
filter	function
filterlist	function
first	function
flush	function
flushlinewise	function
flushsetups	function
flushspacewise	function
~~**get_id**~~	~~function~~
getid	function
getindex	function
id	function
idsplitter	userdata
ifatt	function
ifattempty	function
include	function
inclusion	function
inclusions	function
index	function
inlineverbatim	function
installsetup	function
last	function
lastatt	function
lastmatch	function
lastpar	function
load	function
loadbuffer	function
loaddata	function
loaded	table
main	function
match	function
name	function
namespace	function
nonspace	function
par	function
param	function
parameter	function
path	function
popmatch	function
pos	function
position	function
pure	function

pushmatch	function
raw	function
rawroot	function
rawtex	function
refatt	function
register	function
registerentity	function
removesetup	function
replace	function
resetcdata	function
resetsetups	function
resolvedentity	function
root	function
save	function
serialize	function
setaction	function
setatt	function
setattribute	function
setcdata	function
setcommandtonone	function
setcommandtotext	function
setpar	function
setparam	function
setparameter	function
setsetup	function
snippet	function
sorters	table
F add	
F compare	
F flush	
F reset	
F show	
F sort	
splitid	function
startraw	function
startunescaped	function
stopraw	function
stopunescaped	function
store	function
strip	function
stripped	function
tag	function
texatt	function
text	function
tobuffer	function
tofile	function
toparameters	function
toverbatim	function
verbatim	function
withindex	function
xmltexhandler	data table

math (basic lua)

abs	function
acos	function
asin	function
atan	function
atan2	function
ceil	function
ceiling	function
cos	function
cosd	function
cosh	function
deg	function
div	function
even	function
exp	function
floor	function
fmod	function
huge	number
initialseed	number
ldexp	function
log	function
log10	function
max	function
maxinteger	number
min	function
mininteger	number
mod	function
modf	function
odd	function
pi	number
pow	function
rad	function
random	overloaded function
randomseed	function
round	function
setrandomseedi	function
sin	function
sind	function
sinh	function
sqrt	function
tan	function
tand	function
tanh	function
tointeger	function
type	function
ult	function

mathematics (context)

addfallbacks function
addvariant function
alphabets data table
alternatives data table
beforepassingfonttotex function
big function
boldmap data table
categories table
D functions
charactersets data table
checkaftercopyingfont function
classes data table
classnames data table
codes data table
define function
dictionaries table
D data
D defaults
D groups
F groupset
F groupsetgroup
F name
D names
F registercharacter
F registergroup
F registergroupset
D sets
D variants
dimensions function
domainlabels table
domainnames table
N default
enablecollecting function
extensiblecode function
extensibles data table
extrabase number
fallbacks table
F apply
F install
fallbackstyleattr function
functiontype function
gaps data table
getblobindex function
getboth function
getdomainname function
getmathblob function
getoptionallabel function
getstyle function
gettextblob function
getverboselabel function
horizontalcode function
initializeparameters function
makefont function
mapremap data table
markblobindexdone function
overloadparameters function
preset function
presets table
T less
T more
T zero
privatebase number
registersimplematrix function
registerverboselabel function
remapalphabets function
renderings table
D mappings
D numbers
D sets
renderset function
resetattributes function
scaleparameters function
sets data table
setsnapping function
setvariant function
styles data table
syncboth function
syncname function
syncstyle function
tagging table
B mfenced
toengineclass function

toname function
tostyle function
tweakaftercopyingfont function
tweakbeforecopyingfont function
tweaks table

```
  F accentdimensions
  F action
  F addactuarian
  F addarrows
  F addbars
  F addequals
  F addfourier
  F addmirrors
  F addparts
  F addprivates
  F addrules
  F addscripts
  F bigslots
  F checkaccents
  F checkspacing
  F copyaccents
  T datasets
    T accentdimensions
      T accent
      T over
      T under
    T addarrows
    T addscripts
    T checkspacing
    T copyaccents
    T fixprimes
    T fixslashes
    T wipecues
  F dimensions
  F extendaccents
  F fallbacks
  F fixaccents
  F fixellipses
  F fixintegrals
  F fixoldschool
  F fixprimes
  F fixradicals
  F fixslashes
  F fixvariants
  F flattenaccents
  F inspect
  F keepbases
  F kernpairs
  F kerns
  T kernsets
    T .,
  F margins
  F message
  F moveitalics
  F movelimits
  F parameters
  F radicalbodymargins
  F radicaldegreeanchors
  F replace
  F replacealphabets
  F replacerules
  F setoptions
  F setovershoots
  F showinfo
  F simplifykerns
  F sortvariants
  F staircase
  T subsets
    T acenorsuvxz
    T bhklt
    T d
    T dotless
    T f
    T gjqy
    T horizontalfences
    T i
    T integrals
    T letterlike
    T mw
    T p
  F substitute
  F topanchors
  F tweaks
  F version
  F wipeanchors
  F wipecues
  F wipeitalics
  F wipevariants
```

utfmathaccent	function	**variantcode**	function
utfmathclass	function	**variantcount**	function
utfmathcommand	function	**virtualize**	function
utfmathfiller	function	**virtualized**	data table
utfmathstretch	function		

md5 (extra lua)

HEX	function	**sum**	function
dec	function	**sumHEXA**	function
hex	function	**sumhexa**	function

metapost (context)

checkformat function
checktracingonline function
cm function
codes table

- N addto
- N ampersand
- N and
- N assignment
- N atleast
- N begingroup
- N btex
- N capsule
- N colon
- N comma
- N controls
- N curl
- N cycle
- N definedmacro
- N delimiters
- N endgroup
- N equals
- N etex
- N everyjob
- N exittest
- N expandafter
- N fiorelse
- N if
- N input
- N interim
- N internal
- N iteration
- N leftbrace
- N leftbracket
- N leftdelimiter
- N let
- N macrodef
- N macrospecial
- N maketext
- N message
- N mode
- N newinternal
- N nullary
- N numeric
- N of
- N ofbinary
- N onlyset
- N parametertype
- N pathjoin
- N plusorminus
- N primarybinary
- N primarydef
- N property
- N protection
- N relax
- N repeatloop
- N rightbrace
- N rightbracket
- N rightdelimiter
- N runscript
- N save
- N scantokens
- N secondarybinary
- N secondarydef
- N semicolon
- N setbounds
- N shipout
- N show
- N slash
- N step
- N stop
- N str
- N string
- N tag
- N tension
- N tertiarybinary
- N tertiarydef
- N thingstoadd
- N to
- N typename
- N unary
- N undefinedcs
- N until

N void
N with
N within
N write

colorinitializer	function
convert	function
currentmpx	function
currentmpxstatus	function
defaultformat	string
defaultinstance	string
defaultmethod	string
directrun	function
escaped	function
exectime	table
extensiondata	data table
filterclippath	function
flush	function
flushers	table

T pdf
F flushfigure
F startfigure
F stopfigure
F textfigure
F tocomment

flushnormalpath	function
fonts	table

F define

format_number	function
format_string	function
formatnumber	function
getbendtolerance	function
getbuffer	function
getclippath	function
getextensions	function
getextradata	function
gethashentries	function
getinstancenames	function
getjobdata	function
getmovetolerance	function
getparameter	function
getparameterpreset	function
getparameterset	function
getrecord	function
getstatistics	function
gettext	function
gettolerance	function
graphic	function
graphic_base_pass	function
includesvgbuffer	function
includesvgfile	function
installplugin	function
isobject	function
knownparameters	table
lastlog	string
load	function
maketext	function
metafonts	table
metafun	table

F interpolate
F topath

mptopdf	table

F convertmpstopdf
N nofconverted

n	number
newlogger	function
nofruns	number
nofscriptruns	function
nvformat	function
pdfliterals	function
pluginactions	function
popformat	function
popscriptrunner	function
popvariable	function
popvariables	function
preparetextextsdata	function
process	function
processing	function
processplugins	function
properties	table
pushformat	function
pushscriptrunner	function
pushvariable	function
pushvariables	function
reducetogray	boolean
registerdirect	function
registerscript	function
registertokens	function
remappedtext	function

remaptext	function
reporterror	function
reportsvgfonts	function
reset	function
resetlastlog	function
resetplugins	function
run	function
runinternal	function
runscript	function
scanparameters	function
scripterror	function
scriptindex	function
setbendtolerance	function
setbuffer	function
setextensions	function
setlogging	function
setmovetolerance	function
setoutercolor	function
setparameter	function
setparameterset	function
setrecord	function
settext	function
settolerance	function
setvariable	function
showlog	boolean
showsvgpage	function
simple	function
splitprescript	function
startgraphic	function
startsvghashing	function
stopgraphic	function
stopsvghashing	function
svformat	function
svgcolorremapper	function
svgfactors	table
N %	
N bp	
N cm	
N em	
N ex	
N in	
N mm	
N pc	
N pt	
N px	
svgglyphtomp	function
svgshapestomp	function
svgshapestopdf	function
svgsplitcolor	function
svgtomp	function
svgtopdf	function
sxsy	function
tex	table
F get	
F reset	
F set	
F setfrombuffer	
texerrors	boolean
texexp	function
theclippath	function
tographic	function
totable	function
types	table
N boolean	
N cmykcolor	
N color	
N dependent	
N independent	
N known	
N nep	
N numeric	
N pair	
N path	
N pen	
N picture	
N protodependent	
N string	
N structured	
N suffixedmacro	
N tokenlist	
N transform	
N unknownboolean	
N unknownnep	
N unknownpath	
N unknownpen	
N unknownpicture	
N unknownstring	
N unsuffixedmacro	

N vacuous

typesvgpage	function
unload	function
untagvariable	function
variables	table

mime (context)

b64 function
decode function
decodet table
- F base64
- F quoted-printable

dot function
encode function
encodet table
- F base64
- F quoted-printable

eol function
normalize function
qp function
qpwrp function
report function
stuff function
unb64 function
unqp function
wrap function
wrapt table
- F base64
- F default
- F quoted-printable
- F text

wrp function

mp (context)

autoinject function
aux function
- F boolean
- F color
- F direct
- F draw
- F fill
- F flush
- F fprint
- F integer
- F number
- F numeric
- F pair
- F pairpoints
- F path
- F pathpoints
- F points
- F print
- F quadruple
- F quadruplepoints
- F quoted
- F size
- F string
- F transform
- F triplet
- F tripletpoints
- F vprint

boolean function
cleaned function
color function
dataset function
datasets function
- F load

direct function
draw function
expandtex function
fill function
flatten function
flush function
format function
formatted function
fprint function
get table
- F hashentry

graphformat function
inject function
- F boolean
- F cmykcolor
- F color
- F integer
- F number
- F numeric
- F pair
- F path
- F pt
- F quadruplet
- F string
- F transform
- F triplet
- F whatever
- F whd
- F xy

integer function
lmt_color_functions table
- F shade
- F step

lmt_contours_bitmap_get function
lmt_contours_bitmap_set function
lmt_contours_cleanup function
lmt_contours_color function
lmt_contours_edge_get_band function
lmt_contours_edge_get_cell function
lmt_contours_edge_paths function
lmt_contours_edge_set function
lmt_contours_edge_set_by_band function
lmt_contours_edge_set_by_cell function
lmt_contours_format function
lmt_contours_function function
lmt_contours_maxmean function
lmt_contours_maxz function
lmt_contours_minmean function
lmt_contours_minz function

lmt_contours_nofvalues	function
lmt_contours_nx	function
lmt_contours_ny	function
lmt_contours_range	function
lmt_contours_shade_paths	function
lmt_contours_shade_set	function
lmt_contours_shape_paths	function
lmt_contours_start	function
lmt_contours_stop	function
lmt_contours_value	function
lmt_contours_xrange	function
lmt_contours_yrange	function
lmt_function_x	function
lmt_glyphshape_accent	function
lmt_glyphshape_baseline	function
lmt_glyphshape_boundingbox	function
lmt_glyphshape_depth	function
lmt_glyphshape_height	function
lmt_glyphshape_italic	function
lmt_glyphshape_llx	function
lmt_glyphshape_n	function
lmt_glyphshape_path	function
lmt_glyphshape_start	function
lmt_glyphshape_stop	function
lmt_glyphshape_usedbox	function
lmt_glyphshape_usedline	function
lmt_glyphshape_width	function
lmt_mesh_set	function
lmt_mesh_update	function
lmt_poisson_generate	function
lmt_start_potrace	function
lmt_step_potrace	function
lmt_stop_potrace	function
lmt_surface_do	function
lmt_svg_include	function
mf_blob_add	function
mf_blob_dimensions	function
mf_blob_index	function
mf_blob_new	function
mf_blob_raw_content	function
mf_blob_raw_dimensions	function
mf_blob_raw_reset	function
mf_blob_raw_toutf	function
mf_blob_raw_wipe	function

mf_blob_size	function
mf_blob_width	function
mf_external_figure	function
mf_finish_saving_data	function
mf_formatted_text	function
mf_get_outline_text	function
mf_graphic_text	function
mf_inject_blob	function
mf_map_move	function
mf_map_text	function
mf_named_color	function
mf_outline_text	function
mf_save_data	function
mf_script_index	function
mf_start_saving_data	function
mf_stop_saving_data	function
mode	function
n	function
number	function
numeric	function
pair	function
pairpoints	function
path	function
pathpoints	function
points	function
print	function

```
F boolean
F color
F direct
F draw
F fill
F flush
F fprint
F integer
F number
F numeric
F pair
F pairpoints
F path
F pathpoints
F points
F print
F quadruple
F quadruplepoints
```

F quoted
F size
F string
F transform
F triplet
F tripletpoints
F vprint

processingmode function

quadruple function

quadruplepoints function

quoted function

report function

scan table

F boolean
F cmykcolor
F color
F expression
F integer
F next
F number
F numeric
F pair
F path
F pen
F property
F string
F symbol
F token
T tokenscanners
F transform
T typescanners
F whatever

set table

size function

skip table

F token

solve function

string function

systemmode function

transform function

triplet function

tripletpoints function

vprint function

mplib (extra tex)

execute	function
expandtex	function
finder	function
finders	table
F file	
F mpstemplate	
finish	function
getcallbackstate	function
getcodes	function
getcolormodels	function
getfields	function
gethashentries	function
gethashentry	function
getobjecttypes	function
getstates	function
getstatistics	function
getstatus	function
gettolerance	function
gettype	function
gettypes	function
injectboolean	function
injectcmykcolor	function
injectcolor	function
injectinteger	function
injectnumber	function
injectnumeric	function
injectpair	function
injectpath	function
injectstring	function
injecttransform	function
injectwhatever	function
new	overloaded function
peninfo	function
propertycodes	table
realtimelogging	boolean
scanboolean	function
scancmykcolor	function
scancolor	function
scanexpression	function
scaninteger	function
scannext	function
scannumber	function
scannumeric	function
scanpair	function
scanpath	function
scanpen	function
scanproperty	function
scanstring	function
scansymbol	function
scantoken	function
scantransform	function
settolerance	function
showcontext	function
skiptoken	function
solvepath	function
version	function

noads (context)

handlers table

- F alternates
- F autofences
- F classes
- F collapse
- F dictionaries
- F export
- F families
- F fencing
- F intervals
- F kernpairs
- F makeup
- F normalize
- F numbers
- F relocate
- F render
- F showtree
- F snap
- F spacing
- F suspicious
- F tags
- F unscript
- F unstack
- F variants

process function

processnested function

processors table

- T dictionaries
- T intervals
- F mlisttohlist
- T suspicious
- T unscript
- T unstack

processouter function

node (basic tex)

checkdiscretionaries function
checkdiscretionary function
copy function
copylist function
count function
currentattributes function
dimensions function
direct table

- F addmargins
- F addxoffset
- F addxymargins
- F addyoffset
- F appendaftertail
- F checkdiscretionaries
- F checkdiscretionary
- F collapsing
- F copy
- F copylist
- F copyonly
- F count
- F currentattributes
- F dimensions
- F effectiveglue
- F endofmath
- F exchange
- F fields
- F findattribute
- F findattributerange
- F findnode
- F firstchar
- F firstglyph
- F firstglyphnode
- F firstitalicglyph
- F flattendiscretionaries
- F flattenleaders
- F flushlist
- F flushnode
- F free
- F freeze
- F getanchors
- F getattribute
- F getattributelist
- F getattributes
- F getboth
- F getbottom
- F getbottomdelimiter
- F getbox
- F getcachestate
- F getchar
- F getchardict
- F getcharspec
- F getchoice
- F getclass
- F getcornerkerns
- F getdata
- F getdegree
- F getdelimiter
- F getdenominator
- F getdepth
- F getdirection
- F getdisc
- F getdiscpart
- F getexpansion
- F getfam
- F getfield
- F getfont
- F getgeometry
- F getglue
- F getglyphdata
- F getglyphdimensions
- F getheight
- F getid
- F getindex
- F getinputfields
- F getkern
- F getkerndimension
- F getlanguage
- F getleader
- F getleftdelimiter
- F getlist
- F getlistdimensions
- F getnext

F getnodes
F getnormalizedline
F getnucleus
F getnumerator
F getoffsets
F getoptions
F getorientation
F getparstate
F getpenalty
F getpost
F getpre
F getprev
F getprime
F getpropertiestable
F getproperty
F getreplace
F getrightdelimiter
F getruledimensions
F getscale
F getscales
F getscript
F getshift
F getslant
F getspeciallist
F getstate
F getsub
F getsubpre
F getsubtype
F getsup
F getsuppre
F gettop
F gettopdelimiter
F gettotal
F getusedattributes
F getweight
F getwhd
F getwidth
F getwordrange
F getxscale
F getxyscales
F getyscale
F gluetostring
F hasattribute
F hasdimensions
F hasdiscoption
F hasfield
F hasgeometry
F hasglyph
F hasglyphoption
F hpack
F hyphenating
F id
F ignoremathskip
F insertafter
F insertbefore
F isboth
F ischar
F isdirect
F isglyph
F isitalicglyph
F isnext
F isnextchar
F isnextglyph
F isnode
F isprev
F isprevchar
F isprevglyph
F issimilarglyph
F isspeciallist
F isvalid
F iszeroglue
F kerning
F lastnode
F length
F ligaturing
F makeextensible
F migrate
F mlisttohlist
F naturalhsize
F naturalwidth
F new
F newcontinuationatom
F newmathglyph
F newtextglyph
F patchattributes
F prependbeforehead
F protectglyph
F protectglyphs

F protectglyphsnone
F protrusionskippable
F rangedimensions
F remove
F removefromlist
F repack
F reverse
F serialized
F setanchors
F setattribute
F setattributelist
F setattributes
F setboth
F setbottom
F setbottomdelimiter
F setbox
F setchar
F setchardict
F setchoice
F setclass
F setdata
F setdegree
F setdelimiter
F setdenominator
F setdepth
F setdirection
F setdisc
F setdiscpart
F setexpansion
F setfam
F setfield
F setfont
F setgeometry
F setglue
F setglyphdata
F setheight
F setindex
F setinputfields
F setkern
F setlanguage
F setleader
F setleftdelimiter
F setlink
F setlist
F setnext
F setnucleus
F setnumerator
F setoffsets
F setoptions
F setorientation
F setpenalty
F setpost
F setpre
F setprev
F setprime
F setproperty
F setreplace
F setrightdelimiter
F setruledimensions
F setscales
F setscript
F setshift
F setslant
F setspeciallist
F setsplit
F setstate
F setsub
F setsubpre
F setsubtype
F setsup
F setsuppre
F settop
F settopdelimiter
F settotal
F setweight
F setwhd
F setwidth
F show
F slide
F softenhyphens
F startofpar
F subtypes
F tail
F todirect
F tonode
F tostring
F tovaliddirect
F traverse

F traversechar
F traversecontent
F traverseglyph
F traverseid
F traverseitalic
F traverseleader
F traverselist
F type
F types
F unprotectglyph
F unprotectglyphs
F unsetattribute
F unsetattributes
F usedlist
F usesfont
F verticalbreak
F vpack
F write
F xscaled
F yscaled

effectiveglue	function
endofmath	function
family_font	function
fields	function
findattribute	function
flattendiscretionaries	function
flushlist	function
flushnode	function
free	function
getattribute	function
getcachestate	function
getfield	function
getglue	function
getpropertiestable	function
getproperty	function
gluetostring	function
hasattribute	function
hasfield	function
hasglyph	function
hpack	function
hyphenating	function
id	function
insertafter	function
insertbefore	function
instock	function
inuse	function
isnode	function
iszeroglue	function
kerning	function
lastnode	function
length	function
ligaturing	function
makeextensible	function
mlisttohlist	function
new	function
protectglyph	function
protectglyphs	function
protrusionskippable	function
rangedimensions	function
remove	function
serialized	function
setattribute	function
setfield	function
setglue	function
setproperty	function
show	function
slide	function
subtypes	function
tail	function
todirect	function
tonode	function
tostring	function
traverse	function
traverseid	function
type	function
types	function
unprotectglyph	function
unprotectglyphs	function
unsetattribute	function
usedlist	function
usesfont	function
vpack	function
write	function

nodes (context)

accentcodes	data table
adaptive	table
F handlehorizontal	
F handlevertical	
F set	
aligned	function
append	function
apply	function
applyvisuals	function
astable	function
attributecodes	data table
basepoints	function
boundarycodes	data table
boxes	table
F count	
F dimensions	
F direct	
F dispose	
F found	
F prune	
F reset	
F restore	
F save	
F savenode	
builders	table
F buildpage_filter	
F vbox_filter	
F vpack_filter	
choicecodes	data table
concat	function
copy	function
copy_node	function
copylist	function
countall	function
currentattributes	function
delete	function
destinations	table
N attribute	
T done	
F handler	
T stack	
dircodes	data table
directioncodes	data table
directionvalues	data table
dirvalues	data table
disccodes	data table
discretionarycodes	data table
fencecodes	data table
fields	function
fieldtypes	table
fillcodes	data table
fillvalues	data table
firstdirinbox	function
flush	function
flushlist	function
flushnode	function
fullhpack	function
getattr	function
getattribute	function
getboth	function
getbox	function
getchar	function
getfield	function
getfont	function
getid	function
getleader	function
getlist	function
getnext	function
getprev	function
getproperty	function
getsubtype	function
gluecodes	data table
glyphcodes	data table
handlers	table
F adjusters	
F aligncharacter	
F backgrounds	
F backgroundspage	
F backgroundsvbox	
F characters	
F checkglyphs	
F checkparcounter	

F cleanupbox
F cleanuppage
F export
F finalizebox
F finalizelist
F flatten
F flattenline
F mathalignrelocate
F mathmatrixornaments
F mathmatrixrules
F migrate
F protectglyphs
F recolor
F show
F showhyphenation
F showpreamble
F stripping
F textbackgrounds
F unprotectglyphs
F visualizehyphenation
F wipe

hasattribute function
hasfield function
hpack function
idsandsubtypes function
idstostring function
injections table
F copy
F getligaindex
F getthreshold
F handler
F installgetspaceboth
F installnewkern
F isspace
F keepcounts
F reset
F resetcounts
F setcursive
F setkern
F setligaindex
F setmark
F setmove
F setposition
F setspacekerns

insertafter function
insertbefore function
insertlistafter function
installattributehandler function
is_display_math function
isdirect function
isnode function
isnut function
kerncodes data table
leftmarginwidth function
linefillers table
F filler
F handler
F set

lines table
T boxed
F addlinenumbers
F register
F setup
D data
F finalize
F number

link function
linked function
list function
listcodes data table
listtoutf function
literalvalues data table
locate function
markcodes data table
markers table
F get
F set

mathcodes data table
mathvalues data table
maxboxwidth function
migrations table
F setinserts
F setmarks

new function
noadcodes data table
nodecodes data table
nopts function
nuts table

F addmargins
F addxoffset
F addxymargins
F addyoffset
F append
F appendaftertail
F apply
F applyvisuals
F checkdiscretionaries
F checkslide
F collapsing
T colors
  F set
T components
  F copynocomponents
  F copyonlyglyphs
  F count
  F flush
  F get
  F set
F concat
F copy
F copy_properties
F copylist
F copynode
F copyonly
F copyvisual
F count
F countall
F currentattributes
F delete
F dimensions
F dirdimensions
F effectiveglue
F endofmath
F exchange
F findattribute
F findnode
F firstchar
F firstglyph
F firstglyphnode
F firstitalicglyph
F flattendiscretionaries
F flattenleaders
F flush
F flushlist
F flushnode
F free
F freeze
F fullhpack
F getanchors
F getattr
F getattribute
F getattributelist
F getattributes
F getattrlist
F getattrs
F getboth
F getbottom
F getbottomdelimiter
F getbox
F getboxglue
F getchar
F getchardict
F getcharspec
F getchoice
F getclass
F getcomponents
F getcornerkerns
F getdata
F getdegree
F getdelimiter
F getdenominator
F getdepth
F getdirection
F getdisc
F getdiscpart
F getdiscretionary
F getexpansion
F getfam
F getfield
F getfont
F getgeometry
F getglue
F getglyphdata
F getglyphdimensions
F getheight
F getid

F getindex
F getinputfields
F getkern
F getkerndimension
F getlang
F getlanguage
F getleader
F getleftdelimiter
F getlist
F getlistdimensions
F getmiddle
F getmiddledelimiter
F getnext
F getnormalizedline
F getnucleus
F getnumerator
F getoffsets
F getoptions
F getorientation
F getparstate
F getpenalty
F getpost
F getpre
F getprev
F getprime
F getprop
F getreplace
F getrightdelimiter
F getruledata
F getruledimensions
F getscale
F getscales
F getscript
F getshift
F getslant
F getspeciallist
F getstate
F getsub
F getsubpre
F getsubtype
F getsup
F getsuppre
F getsurround
F gettop
F gettopdelimiter
F gettotal
F getusedattributes
F getvalue
F getweight
F getwhd
F getwidth
F getwordrange
F getxscale
F getxyscales
F getyscale
F gluetostring
F hasattribute
F hasdimensions
F hasdiscoption
F hasfield
F hasgeometry
F hasglyph
F hasglyphoption
F hpack
F hyphenating
F ignoremathskip
F insertafter
F insertbefore
F insertlistafter
F is_display_math
F isboth
F ischar
F isdirect
F isdone
F isglyph
F isitalicglyph
F isnext
F isnextchar
F isnextglyph
F isnode
F isnut
F isprev
F isprevchar
F isprevglyph
F issimilarglyph
F iszeroglue
F kerning
F lastnode

F length
F ligaturing
F link
F linked
F locate
F migrate
F migratebox
F mlisttohlist
F naturalhsize
F naturalwidth
F nestedtracedslide
F new
F newcontinuationatom
F newmathglyph
F openup
F patchattributes
T pool
  F accent
  F baselineskip
  F boundary
  F boxrule
  F choice
  F cleanup
  F delim
  F delimiter
  F direction
  F directliteral
  F disc
  F emptyrule
  F fence
  F fontkern
  F fraction
  F glue
  F gluespec
  F glyph
  F hlist
  F imagerule
  F indentskip
  F italickern
  F kern
  F latelua
  F leader
  F lefthangskip
  F leftmarginkern
  F leftskip
  F lineskip
  F literal
  F mathchar
  F mathtextchar
  F negatedglue
  F noad
  F originliteral
  F outlinerule
  F pageliteral
  F penalty
  F radical
  F rawliteral
  F register
  F restore
  F righthangskip
  F rightmarginkern
  F rightskip
  F rule
  F save
  F savepos
  F setmatrix
  F setstate
  F shrink
  F spacefontkern
  F startrotation
  F stock
  F stretch
  F strutrule
  F subbox
  F submlist
  F temp
  F usage
  F userdefined
  T userids
  F usernode
  F userrule
  F virtualrule
  F vlist
  F wordboundary
F prepend
F prependbeforehead
F processranges
F processwords

F protectglyph
F protectglyphs
F protectglyphsnone
F protrusionskippable
F rangedimensions
F rawprop
F reference
F rehpack
F remove
F removefromlist
F repack
F repackhlist
F replace
F reverse
T rules
  T ruleactions
    F box
    F draw
    F fill
    F mp
    F stroke
  F userrule
F serialized
F setanchors
F setattr
F setattribute
F setattributelist
F setattributes
F setattrlist
F setattrs
F setboth
F setbottom
F setbox
F setboxglue
F setchar
F setchardict
F setchoice
F setclass
F setcomponents
F setdata
F setdegree
F setdelimiter
F setdenominator
F setdepth
F setdirection
F setdisc
F setdiscpart
F setdiscretionary
F setexpansion
F setfam
F setfield
F setfont
F setgeometry
F setglue
F setglyphdata
F setheight
F setinputfields
F setkern
F setlang
F setlanguage
F setleader
F setleftdelimiter
F setlink
F setlist
F setmiddledelimiter
F setnext
F setnucleus
F setnumerator
F setoffsets
F setoptions
F setorientation
F setparproperty
F setpenalty
F setpost
F setpre
F setprev
F setprime
F setprop
F setreplace
F setrightdelimiter
F setruledata
F setruledimensions
F setscale
F setscales
F setscript
F setshift
F setslant
F setspeciallist

F setsplit
F setstate
F setsub
F setsubpre
F setsubtype
F setsup
F setsuppre
F setsurround
F settop
F setvalue
F setvisual
F setvisuals
F setweight
F setwhd
F setwidth
F show
F slide
F softenhyphens
F somepenalty
F somespace
F splitbox
F startofpar
F striprange
F tail
F takeattr
F takebox
F takelist
F theprop
F tonode
F tonodes
F tonut
F tosequence
F tostring
F tracedslide
F traverse
F traversechar
F traversecontent
F traverseglyph
F traverseid
F traverseitalic
F traverseleader
F traverselist
T traversers
  F alignrecord
  F boundary
  F char
  F content
  F dir
  F disc
  F glue
  F glyph
  F hlist
  F leader
  F list
  F mark
  F next
  F node
  F penalty
  F rule
  F unset
  F vlist
  F whatsit
T treversers
  F char
  F content
  F glyph
  F leader
  F list
  F node
T typesetters
  F tohbox
  F tohpack
  F tonodes
  F tovbox
  F tovpack
F unprotectglyph
F unprotectglyphs
F unsetattribute
F unsetattributes
F untracedslide
F usedlist
F usesfont
F verticalbreak
F vianodes
F vianuts
F vpack
F write
F xscaled

F yscaled

packlist function

parcodes data table

penaltycodes data table

plugindata table

F finalizer

F initializer

S name

T namespace

N attribute

D data

F enable

F handler

F register

D registered

D values

F processor

points function

pool table

F baselineskip

F cleanup

F directliteral

F glyph

F hlist

F latelua

F leader

F lineskip

F literal

F originliteral

F register

F restore

F save

F savepos

F setmatrix

F stock

F usage

F usernode

F vlist

prepend function

print function

processors table

F alignment_filter

F append_line_filter

F glyph_run

F post_linebreak_filter

F pre_linebreak_filter

F trace_glyph_run_a

F trace_glyph_run_b

properties table

T data

pts function

radicalcodes data table

references table

N attribute

T done

F handler

T stack

rehpack function

remove function

repackhlist function

replace function

report function

rightmarginwidth function

rulecodes data table

rules table

F handler

F process

F set

F userrule

serialize function

serializebox function

serialized function

setattr function

setattribute function

setboth function

setbox function

setboxtonaturalwd function

setchar function

setfield function

setfont function

setleader function

setlink function

setlist function

setnext function

setprev function

setproperty function

shifts table

F handler

F set

showboxes function

showlist function

showsimplelist function

skipcodes data table

snapshots table

F getsamples

F resetsamples

F takesample

somepenalty function

somespace function

specialskipcodes table

splitbox function

startofpar function

stripdiscretionaries function

stripping table

T glyphs

F set

subtypes data table

tail function

takeattr function

takebox function

tasks table

F actions

F appendaction

F disableaction

F disablegroup

F enableaction

F enablegroup

F freeze

F freezegroup

F new

F prependaction

F removeaction

F replaceaction

F restart

F setaction

F showactions

F table

tobasepoints function

tocentimeters function

tociceros function

todidots function

todimen function

toinches function

tomillimeters function

tonode function

tonodes function

tonut function

topicas function

topoints function

toscaledpoints function

tosequence function

tostring function

totable function

totree function

toutf function

tracers table

T colors

F reset

F set

F setlist

F directions

F fontchar

T frequencies

F register

T pool

T nodes

F rule

T nuts

F rule

F resetproperties

F rule

F setlist

F setproperties

F start

T steppers

F check

F codes

F features

F font

F glyphs

F message

F messages

F nofsteps

F register

F reset

F start

F stop
F stop
T transparencies
F reset
F set
F setlist
traverse function
traverseid function
typesetters table
F hbox
F hpack
F tohbox
F tohpack
F tonodes
F tovbox
F tovpack
F vpack
unsetattribute function
usedlist function
vianodes function
vianuts function
visualizebox function
visualizers table
F attribute
T attributes
F box
F definelayer
F enable
T filters
N hglue
N hkern
N hpenalty
N vglue
N vkern
N vpenalty
F getusedfont
F handler
F markfonts
T modes
N boundary
N depth
N dir
N discretionary
N expansion
N fontkern
N glue
N glyph
N hbox
N hglue
N hkern
N hpenalty
N insert
N italic
N kern
N line
N marginkern
N mark
N math
N mathglue
N mathkern
N origin
N par
N penalty
N simple
N simplehbox
N simplevbox
N simplevtop
N space
N strut
N user
N vbox
N vglue
N vkern
N vpenalty
N vtop
N whatsit
F register
F registercache
F setfont
F setlayer
F setruledglyph
F setvisual
vpack function
whatsitcodes data table
write function

number (context)

	function
basepoints	function
bytetodecimal	function
compactfloat	function
decimaltobyte	function
dimenfactors	data table
formatted	function
formattedfloat	function
idiv	function
maxdimen	number
nopts	function
nubasepoints	function
nupoints	function
percent	function
points	function
pts	function
signed	function
sparseexponent	function
splitdimen	function
to16dot16	function
tobasepoints	function
tobitstring	function
tocentimeters	function
tociceros	function
tocontext	function
todidots	function
todimen	function
toediths	function
toevenhex	function
toinches	function
tomillimeters	function
tonumberf	function
tonumberg	function
topicas	function
topoints	function
toscaled	function
toscaledpoints	function
totoves	function
valid	function

objects (context)

data	table	**page**	function
dimensions	function	**reference**	function
found	function	**register**	function
n	number	**restore**	function

optional (context)

curl table
- F escape
- F fetch
- F getversion
- F initialize
- F unescape

foreign table
- F abivalues
- F getbuffer
- F initialize
- F load
- F newbuffer
- F totable
- F types

ghostscript table
- F execute
- F initialize

graphicsmagick table
- F execute
- F initialize

hb table
- F getshapers
- F getversion
- F initialize
- F loadfont
- F shapestring

imagemagick table
- F execute
- F initialize

kpse table
- F expand_braces
- F expand_path
- F expand_var
- F find_file
- F find_files
- F get_file_types
- F initialize
- F readable_file
- F set_program_name
- F var_value

library table
- F load

loaded table

lz4 table
- F compress
- F decompresssize
- F framecompress
- F framedecompress
- F initialize

lzma table
- F compress
- F decompress
- F initialize

lzo table
- F compress
- F decompresssize
- F initialize

mujs table
- F dofile
- F execute
- F initialize
- F reset
- F setclosefile
- F setconsole
- F setfindfile
- F setopenfile
- F setreadfile
- F setseekfile

mysql table
- F close
- F execute
- F getmessage
- F initialize
- F open

openssl table
- F getversion
- F initialize
- F sign
- F verify

postgress table
- F close
- F execute

F getmessage

F initialize

F open

sqlite table

F close

F execute

F getmessage

F initialize

F open

zint table

F execute

F initialize

zstd table

F compress

F decompress

F initialize

os (basic lua)

__getenv__	function
__setenv__	function
binsuffix	string
binsuffixes	table
bits	number
clock	function
converttime	function
date	overloaded function
difftime	function
enableansi	function
env	data table
execute	overloaded function
exit	overloaded function
fulltime	function
getcodepage	function
getenv	overloaded function
gettimeofday	function
isleapyear	function
launch	function
libsuffix	string
localtime	function
name	string
newline	string
nofdays	function
now	function
pipeto	function
platform	string
remove	overloaded function
rename	overloaded function
resultof	function
runtime	function
selfarg	table
selfbin	string
selfcore	string
selfdir	string
selflink	string
selfname	string
selfpath	string
setenv	overloaded function
setexitcode	function
setlocale	overloaded function
sleep	function
time	function
timezone	function
tmpname	function
today	function
type	string
uname	function
uuid	function
validdate	function
weekday	function
where	function
which	function

package (basic lua)

config	string
cpath	string
extralibpath	function
extralibpaths	function
extraluapath	function
extraluapaths	function
helpers	table
libpaths	function
loaded	table
loaders	table
loadlib	function
luapaths	function
path	string
poplibpath	function
popluapath	function
preload	table
pushlibpath	function
pushluapath	function
searchers	table
searchpath	function

pagebuilders (context)

columnsets table

- F add
- F block
- F check
- F currentcolumn
- F define
- F finishflush
- F flushareas
- F flushcolumn
- F flushrest
- F goto
- F noto
- F prepareflush
- F put
- F registerarea
- F reset
- F setarea
- F sethsize
- F sethspan
- F setlines
- F setproperties
- F setstart
- F setvsize

injections table

- F flushafter
- F flushbefore
- F save

mixedcolumns table

- F cleanup
- F finalize
- F getlist
- F getrest
- F getsplit
- F setsplit

pdf (context)

immediateobj function

pdfe (extra tex)

arraytotable	function
close	function
closestream	function
dictionarytotable	function
getarray	function
getboolean	function
getbox	function
getcatalog	function
getdictionary	function
getfromarray	function
getfromdictionary	function
getfromobject	function
getfromreference	function
getfromstream	function
getinfo	function
getinteger	function
getmemoryusage	function
getname	function
getnofobjects	function
getnofpages	function
getnumber	function
getobjectrange	function
getpage	function
getpages	function
getsize	function
getstatus	function
getstream	function
getstring	function
gettrailer	function
getversion	function
new	function
open	function
openfile	function
openstream	function
pagestotable	function
readfromstream	function
readwholestream	function
type	function
unencrypt	function

physics (context)

units table

F registershortcut

F registerunit

D tables

pngdecode (extra tex)

applyfilter	function
expand	function
interlace	function
makemask	function
splitmask	function
tocmyk	function
tomask	function

posit (extra lua)

NaN function
NaR function
__add function
__band function
__bor function
__bxor function
__div function
__eq function
__index table

```
  F NaN
  F NaR
  F __add
  F __band
  F __bor
  F __bxor
  F __div
  F __eq
  T __index
    F NaN
    F NaR
    F __add
    F __band
    F __bor
    F __bxor
    F __div
    F __eq
    F __le
    F __lt
    F __mul
    S __name
    F __pow
    F __shl
    F __shr
    F __sub
    F __tostring
    F __unm
    F abs
    F acos
    F asin
    F atan
    F band
    F bor
    F bxor
    F ceil
    F conj
    F copy
    F cos
    F exp
    F exp2
    F floor
    F fromposit
    F integer
    F log
    F log10
    F log1p
    F log2
    F logb
    F max
    F min
    F modf
    F new
    F pow
    F rotate
    F round
    F rounded
    F shift
    F sin
    F sqrt
    F tan
    F tonumber
    F toposit
    F tostring
  F __le
  F __lt
  F __mul
  S __name
  F __pow
  F __shl
  F __shr
  F __sub
  F __tostring
  F __unm
```

```
  F abs
  F acos
  F asin
  F atan
  F band
  F bor
  F bxor
  F ceil
  F conj
  F copy
  F cos
  F exp
  F exp2
  F floor
  F fromposit
  F integer
  F log
  F log10
  F log1p
  F log2
  F logb
  F max
  F min
  F modf
  F new
  F pow
  F rotate
  F round
  F rounded
  F shift
  F sin
  F sqrt
  F tan
  F tonumber
  F toposit
  F tostring
__le        function
__lt        function
__mul       function
__name      string
__pow       function
__shl       function
__shr       function
__sub       function
__tostring  function
__unm       function
abs         function
acos        function
asin        function
atan        function
band        function
bor         function
bxor        function
ceil        function
conj        function
copy        function
cos         function
exp         function
exp2        function
floor       function
fromposit   function
integer     function
log         function
log10       function
log1p       function
log2        function
logb        function
max         function
min         function
modf        function
new         function
pow         function
rotate      function
round       function
rounded     function
shift       function
sin         function
sqrt        function
tan         function
tonumber    function
toposit     function
tostring    function
```

potrace (extra tex)

checked	function
concat	function
contourplot	function
convert	function
count	function
flush	function
free	function
getbitmap	function
new	function
process	function
setbitmap	function
stripped	function
totable	function
trace	function

publications (context)

addtexentry function
analyze function
anywhere function
authorcache data table
authorcollapsers data table
authorhashers table
- F invertedshort
- F normal
- F normalinverted
- F normalshort
- F sequential
- F sequentialshort
- F short

authormap data table
authortoregister function
btx table
casters table
- F author
- F keyword
- F pagenumber
- F range

checkeddataset function
citevariants table
- F author
- F authornum
- F authoryear
- F authoryears
- F category
- F default
- F entry
- F index
- F keywords
- F num
- F page
- F short
- F tag
- F textnum
- F year

collected data table
components table
- F author

concatstate function
converttoxml function
currentspecification table
- S author
- T categories
  - T article
    - S category
    - T extra
    - T fields
      - S author
      - S editor
      - S journal
      - S keywords
      - S note
      - S number
      - S pages
      - S title
      - S volume
      - S year
    - T optional
    - T required
    - T sets
      - T author
    - T virtual
  - T book
    - S category
    - T extra
    - T fields
      - S address
      - S author
      - S edition
      - S editor
      - S keywords
      - S note
      - S number
      - S publisher
      - S subtitle
      - S title
      - S volume
      - S year
    - T optional

```
    T required
    T sets
      T author
      T editionset
    T virtual
  T demo-a
    S category
    T extra
    T fields
      S author
      S institution
      S organization
      S subtitle
      S title
      S year
    T optional
    T required
    T sets
      T author
    T virtual
  T demo-b
    S category
    T extra
    T fields
      S author
      S institution
      S organization
      S subtitle
      S title
      S year
    T optional
    T required
    T sets
      T authors
    T virtual
S comment
S copyright
T fields
S name
T types
  S author
  S editor
  S keywords
  S number
    S page
    S pages
    S volume
  S version
  D virtual
  T virtualfields
    B authornum
    B authoryear
    B authoryears
    B num
    B suffix
```

datasets	table
detailed	table
endnotes_to_btx	function
enhancer	table

```
  S arguments
  T askip
  B dirty
  T gskip
  T kind
  T list
    T system
  T order
  N steps
```

enhancers	table

```
  F details
  F order
  F suffixes
```

finder	function
firstofrange	function
found	function
get	function
getcasted	function
getdetail	function
getdirect	function
getfaster	function
getfield	function
getfuzzy	function
getindex	function
inspectors	data table
journals	table

```
  F abbreviated
  F add
  F expanded
```

F load
F save
lastofrange function
lists table
F collectentries
F combiinlist
F fetchentries
F filterall
F flushcombi
F flushentry
F flushtag
T methods
F dataset
F force
F global
F local
F nofregistered
F prepareentries
F register
F sameasprevious
T sorters
F cite
F dataset
F default
F index
F list
F reference
F short
F used
F uservariable
listvariants table
F default
F index
F num
F short
F tag
load function
loadbibdata function
loaddefinitionfile function
loaders table
F bbl
F bib
F buffer
F endnote
F json
F lua
F refman
F registercleaner
F xml
loadreplacementfile function
new function
okay function
oneorrange function
parenttag function
permitted function
prerollcmdstring function
refman_to_btx function
resolvedname function
save function
savers table
F bib
F lua
F xml
search function
setcategory function
setspecification function
singularorplural function
sortedentries function
sortmethods table
T authortitle
T sequence
T authoryear
T sequence
T dataset
T sequence
T key
T sequence
specifications table
T default
S author
S comment
S copyright
S name
S version
D virtual
splitauthor function
splitoneauthor function
statistics table

N nofbytes
N nofdatasets
N nofdefinitions
N nofshortcuts

tables table
D implicits
D origins
D privates
D sets
D specials
D types
D virtuals

tags function

tracers table
F showdatasetauthors
F showdatasetcompleteness
F showdatasetfields
F showentry
F showfields
F showhashedauthors
F showtables
F showvariants

typesetters table
F author
F string

usedataset function

usedentries function

writers table
F author
F keyword
F pagenumber
F range

regimes (context)

backmapping	data table	**mapping**	data table
cleanup	function	**pop**	function
disable	function	**process**	function
enable	function	**push**	function
fromregime	function	**toregime**	function
list	function	**translate**	function

resolvers (context)

allprefixes	function
appendhash	function
automount	function
booleanvariable	function
cacheversion	string
checkedvariable	function
cleanedpathlist	function
cleaners	data table
cleanpath	function
cleanpathlist	function
cleanupbinfile	function
concatinators	data table
configbanner	string
configurationfiles	function
dangerous	data table
datastate	function
decompressors	table
F register	
F unregister	
defaultsuffixes	table
dirstatus	function
dowithfilesandreport	function
dowithfilesintree	function
dowithpath	function
dowithvariable	function
executers	table
expandbraces	function
expandedpathfromlist	function
expandedpathlist	function
expandedpathlistfromvariable	function
expandpath	function
expandpathfromvariable	function
expansion	function
extendtexmfvariable	function
filtered_from_content	function
find_file	function
find_files	function
findbinfile	function
findctxfile	function
finders	data table
findfile	function
findfiles	function
findgivenfile	function
findgivenfiles	function
findlib	function
findpath	function
findpk	function
findtexfile	function
findwildcardfile	function
findwildcardfiles	function
formatofsuffix	function
formatofvariable	function
formats	data table
foundintree	function
foundintrees	function
generators	data table
get_from_content	function
getenv	function
getextrapaths	function
getfilehashes	function
gethashes	function
getownpath	function
getreadfilename	function
hashers	data table
homedir	string
ignorelibrary	function
initialized	function
installinputfilehandler	function
installinputlinehandler	function
jobpath	function
jobs	table
F collectstructure	
F currentcomponent	
F currentenvironment	
F currentproduct	
F currentproject	
F currentstructure	
F justacomponent	
F namelist	
D processors	
F productcomponent	
F usefile	

joinpath	function
knownvariables	function
libraries	table
F optionalloaded	
F validoptional	
load	function
loadbinfile	function
loaders	data table
loadlualib	function
loadtexfile	function
loadtime	function
locateformat	function
locators	data table
luacnffallback	string
luacnfname	string
luacnfspec	string
luacnfstate	string
macros	table
F convertfile	
F decodecomment	
F enablecomment	
F encodecomment	
F preprocessed	
F processmk	
F processmklx	
F processmkvi	
F version	
maxreadlevel	number
methodhandler	function
openbinfile	function
openers	data table
openfile	function
opentexfile	function
ownpath	string
popextrapath	function
popinputname	function
poppath	function
prefixes	data table
prependhash	function
pushextrapath	function
pushinputname	function
pushpath	function
registerextrapath	function
registerfile	function
registerfilehash	function
registerintrees	function
registermethod	function
registerzipfile	function
relations	data table
renew	function
renewcache	function
repath	function
reset	function
resetextrapaths	function
resetresolve	function
resolve	function
savers	data table
scandata	function
scanfiles	function
schemes	table
T cleaners	
F md5	
F none	
F strip	
F cleanname	
F fetchstring	
F fetchtable	
F install	
setdynamic	function
setenv	function
showpath	function
simplescanfiles	function
splitconfigurationpath	function
splitexpansions	function
splitmethod	function
splitpath	function
stackpath	function
starttiming	function
stoptiming	function
suffixes	data table
suffixmap	data table
suffixofformat	function
texdatablob	function
toppath	function
tracers	data table
unexpandedpath	function
unexpandedpathlist	function
unresolve	function

updaterelations	function	**validatedpath**	function
updatescript	function	**variable**	function
uselibrary	function	**variableofformat**	function
usertypes	data table	**variableofformatorsuffix**	function
usezipfile	function	**wildcardpattern**	function

sandbox (context)

disablelibraries	function
disablerunners	function
enable	function
filehandlerone	function
filehandlertwo	function
finalizer	function
getrunner	function
initializer	function
iohandler	function
logcalls	function
logfiles	function
original	function
overload	function
possiblepath	function
redefine	function
register	function
registerbinary	function
registerlibrary	function
registerroot	function
registerrunner	function
report	function
setfilenamelogger	function
setlogger	function
setreporter	function
settrace	function
validfilename	function

scripts (context)

autofontfeature table
- F disable
- F enable
- F handler

colors data table
decomposehangul function
defaults table
- N inter_char_half_shrink_factor
- N inter_char_half_stretch_factor
- N inter_char_hangul_penalty
- N inter_char_quarter_shrink_factor
- N inter_char_quarter_stretch_factor
- N inter_char_shrink_factor
- N inter_char_stretch_factor
- N inter_word_stretch_factor

getdata function
getinjector function
getsplitter function
getstatus function
handlers data table
helpers data table
injectors data table
inserters table
- F nobreakspace_after
- F nobreakspace_before
- F space_after
- F space_before
- F zerowidthspace_after
- F zerowidthspace_before

installdataset function
installmethod function
numbertodataset data table
numbertohandler data table
reset function
set function
setstatus function
splitters data table

set (context)

contains function
create function
tolist function
tonumber function
totable function

sha2 (extra lua)

HASH256	function	**hash256**	function
HASH384	function	**hash384**	function
HASH512	function	**hash512**	function
HEX256	function	**hex256**	function
HEX384	function	**hex384**	function
HEX512	function	**hex512**	function
digest256	function	**sum256**	function
digest384	function	**sum384**	function
digest512	function	**sum512**	function

sio (extra lua)

read2dot14	function
readbytes	function
readbytetable	function
readcardinal1	function
readcardinal1le	function
readcardinal2	function
readcardinal2le	function
readcardinal3	function
readcardinal3le	function
readcardinal4	function
readcardinal4le	function
readcardinaltable	function
readcline	function
readcstring	function
readfixed2	function
readfixed4	function
readinteger1	function
readinteger1le	function
readinteger2	function
readinteger2le	function
readinteger3	function
readinteger3le	function
readinteger4	function
readinteger4le	function
readintegertable	function
tocardinal1	function
tocardinal1le	function
tocardinal2	function
tocardinal2le	function
tocardinal3	function
tocardinal3le	function
tocardinal4	function
tocardinal4le	function

socket (extra lua)

BLOCKSIZE	number
_DATAGRAMSIZE	number
_SETSIZE	number
_SOCKETINVALID	number
bind	function
choose	function
connect	function
connect4	function
connect6	function
dns	table

```
  F getaddrinfo
  F gethostname
  F getnameinfo
  F tohostname
  F toip
```

gettime	function
headers	table

```
  T canonic
    S accept
    S accept-charset
    S accept-encoding
    S accept-language
    S accept-ranges
    S action
    S age
    S allow
    S alternate-recipient
    S arrival-date
    S authorization
    S bcc
    S cache-control
    S cc
    S comments
    S connection
    S content-description
    S content-disposition
    S content-encoding
    S content-id
    S content-language
    S content-length
    S content-location
    S content-md5
    S content-range
    S content-transfer-encoding
    S content-type
    S cookie
    S date
    S diagnostic-code
    S dsn-gateway
    S etag
    S expect
    S expires
    S final-log-id
    S final-recipient
    S from
    S host
    S if-match
    S if-modified-since
    S if-none-match
    S if-range
    S if-unmodified-since
    S in-reply-to
    S keywords
    S last-attempt-date
    S last-modified
    S location
    S max-forwards
    S message-id
    S mime-version
    S original-envelope-id
    S original-recipient
    S pragma
    S proxy-authenticate
    S proxy-authorization
    S range
    S received
    S received-from-mta
    S references
    S referer
    S remote-mta
    S reply-to
    S reporting-mta
```

S resent-bcc
S resent-cc
S resent-date
S resent-from
S resent-message-id
S resent-reply-to
S resent-sender
S resent-to
S retry-after
S return-path
S sender
S server
S smtp-remote-recipient
S status
S subject
S te
S to
S trailer
S transfer-encoding
S upgrade
S user-agent
S vary
S via
S warning
S will-retry-until
S www-authenticate
S x-mailer
F lower
F normalize

http table
N TIMEOUT
S USERAGENT
F genericform
F open
F request

newtry function

protect function

report function

select function

sink function

sinkt table
F close-when-done
F default
F http-chunked
F keep-open

skip function

sleep function

source function

sourcet table
F by-length
F default
F http-chunked
F until-closed

tcp function

tcp4 function

tcp6 function

tp table
N TIMEOUT
F connect

try function

udp function

udp4 function

udp6 function

url table
F absolute
F build
F build_path
F escape
F parse
F parse_path
F unescape

sorters (context)

basicsorter function

comparers table

F basic

constants table

S defaultdigits

S defaultlanguage

S defaultmethod

N digitsmaximum

N digitsoffset

N ignoredoffset

N replacementoffset

T validmethods

B ch

B mc

B mm

B pc

B pm

B uc

B zc

B zm

definitions data table

firstofsplit function

packch function

packuc function

replacementlist function

setlanguage function

sort function

splitters table

F utf

strip function

tracers data table

update function

sparse (extra lua)

concat	function	**range**	function
get	function	**restore**	function
max	function	**set**	function
min	function	**traverse**	function
new	function	**wipe**	function

statistics (context)

benchmarktimer	function
checkfmtstatus	function
currenttime	function
elapsed	function
elapsedindeed	function
elapsedseconds	function
elapsedtime	function
enable	boolean
formatruntime	function
hastiming	function
memused	function
register	function
resettiming	function
runtime	function
savefmtstatus	function
show	function
showmemory	function
showusage	function
starttiming	function
stoptiming	function
threshold	number
timed	function
tracefunction	function

status (basic tex)

getbufferstate	function
getcallbackstate	function
getconstants	function
geterrorlinestate	function
geterrorstate	function
getexpandstate	function
getextrastate	function
getfilestate	function
getfontstate	function
gethalferrorlinestate	function
gethashstate	function
getinputstate	function
getinsertstate	function
getlanguagestate	function
getlookupstate	function
getluastate	function
getmarkstate	function
getneststate	function
getnodestate	function
getparameterstate	function
getpoolstate	function
getreadstate	function
getsavestate	function
getsparsestate	function
getstringstate	function
gettexstate	function
gettokenstate	function
getwarningstate	function
iocodes	table
list	function
resetmessages	function

storage (context)

allocate	function	**noftables**	number
data	table	**register**	function
mark	function	**shared**	data table
marked	function	**storedfunctions**	data table
max	number	**strip**	boolean
min	number	**tofmodules**	number
nofmodules	number	**toftables**	number

streams (context)

collect	function
disable	function
enable	function
flush	function
initialize	function
push	function
start	function
stop	function
synchronize	function

string (basic lua)

autodouble	function
autosingle	function
booleanstring	function
byte	function
bytepairs	function
bytes	function
bytetable	function
char	function
characterpairs	function
characters	function
checkedsplit	function
chrtointeger	function
collapsespaces	function
containsws	function
count	function
dawson	function
dectointeger	function
dump	function
erf	function
erfc	function
erfcx	function
erfi	function
escapedpattern	function
escapedquotes	function
explode	function
f6	function
find	function
format	function
formatter	function
formatters	function
fullstrip	function
gmatch	function
gsub	function
hextocharacters	function
hextointeger	function
is_boolean	function
is_empty	function
itself	function
len	function
limit	function
linetable	function
longtostring	function
lower	function
match	function
nospaces	function
nspaces	table
octtointeger	function
optionalquoted	function
pack	function
packrowscolumns	function
packsize	function
quote	function
quoted	function
rep	function
replacenewlines	function
reverse	function
split	function
splitlines	function
splitup	function
strip	function
sub	function
texhashed	function
texnewlines	function
tformat	function
toHEX	function
toboolean	function
tobytes	function
tocontext	function
todec	function
todimen	function
tohex	function
tonodes	function
topattern	function
totable	function
tounicode16	function
toutf	function
toutf32	function
toutf8	function
tracedchar	function
tracedchars	table
unescapedquotes	function
unpack	function

unquote	function
unquoted	function
upper	function
utfcharacter	function
utfcharacters	function
utfcharactertable	function
utflength	function
utfpadd	function
utfpadding	function
utfvalue	function
utfvalues	function
utfvaluetable	function
valid	function
voigt	function
voigt_hwhm	function
wordsplitter	function

structures (context)

atlocation function
blocks table
- D collected
- F define
- F empty
- F print
- F save
- F select
- F setstate
- D states
- D tobesaved
- F uservariable

bookmarks table
- T extras
  - F get
  - F mergesections
  - F register
  - F reset
  - F tosections
- F finalize
- F flatten
- F installhandler
- F merge
- S method
- F overload
- F place
- F register
- F setopened
- F setup

counters table
- F add
- F analyze
- F check
- D collected
- F compact
- F converted
- F currentvalue
- D data
- F define
- F first
- F get
- F last
- F next
- F prev
- F previous
- F raw
- F record
- F reset
- F restart
- F restore
- F save
- F set
- F setlevel
- F setoffset
- F setown
- F setstate
- F setvalue
- T specials
  - F userpage
- F subs
- D tobesaved
- F value
- F way

currentsectionnumber function
descriptions table
documents table
- D data
- F initialize
- F preset
- F reset

floats table
formulas table
helpers table
- F analyze
- F formulanumber
- F merged
- F page
- F pageofinternal
- F prefix
- F prefixlastpage
- F prefixpage
- F simplify

F title
F touserdata

inserts table

D data
F define
F getdata
F getlocation
F getname
F getnumber
F setlocation
F setup
D stored

itemgroups table

D collected
D tobesaved

lists table

F addto
F analyze
B autoreorder
D cached
D collected
F discard
F enhance
T enhancers
F external
F filter
F formulanumber
F getresult
F groupindex
F hasnumberdata
F haspagedata
F hastitledata
F integrate
F iscached
D kinds
F label
F location
D names
F number
D ordered
F page
F popnesting
F prefix
F prefixednumber
F prefixedpage
F process
D pushed
F pushnesting
F rawnumber
F realpage
F reordered
T result
F savedlistnumber
F savedlistprefixednumber
F savedlisttitle
D sectionblocks
F sectionnumber
F size
F title
D tobesaved
F use
F userdata
F uservalue

marks table

F number
F title

notes table

F check_spacing
F checkpagechange
F define
F doifonsamepageasprevious
F get
F getdeltapage
F getinternal
F getn
F getstate
F hascontent
F internal
F internalid
F listindex
D numbers
F ordered
F postpone
F resetpostponed
F restore
F save
F setstate
D states

F store

pages table

F allmarked
F analyze
D collected
F fraction
F in_body
F is_odd
F mark
F marked
F markedlist
N nofpages
F number
F on_right
F save
D tobesaved
F toranges

references table

F analyze
F checked
F checkedfile
F checkedfileorurl
F checkedprogram
F checkedrealpage
F checkedurl
D collected
U componentsplitter
F currentmetadata
F currentorder
D currentset
F define
D defined
D derived
F enhance
D executers
F export
T files
  T data
  F define
  F found
  F get
F filter
F filterdefault
D filters
D flaginternals
F from
D functions
F get
F getcurrentmetadata
F getcurrentprefixspec
F getinnermethod
F getinternalcachedlistreference
F getinternallistreference
F getinternalreference
F getoutermethod
D handlers
F identify
F import
F inject
F injectcurrentset
S innermethod
F integrate
D internals
F load
F loadpresets
F mark
S outermethod
F pagefromhash
D pages
U prefixsplitter
D productdata
T programs
  T data
  F define
  F get
F realpageofpage
U referencesplitter
D referred
F referredpage
F registerfinalizer
F registerinitializer
F registerpage
F reset
D runners
F set
F setandgetattribute
F setinternalreference
F setlinkmethod

F setnextinternal
F setnextorder
F setreferencerealpage
F setstructure
D specials
F splitcomponent
F splitprefix
F splitreference
F synchronizepage
T testrunners
  F inner
  F special
  F special operation
  F special operation with arguments
T testspecials
  F deltapage
  F i
  F internal
  F page
  F realpage
  F section
  F userpage
D tobereferred
D tobesaved
D tracedpages
T urls
  T data
  F define
  F found
  F get
  F untex
D usedinternals
D usedviews
F valid

registers table

F analyze
D collected
F compare
F define
F enhance
F extend
F filter
F filtercollected
F finalize
F findinternal
F flush
F get
F integrate
F prepare
F process
F setmethod
F sort
F store
D tobesaved
F tweak
F unique
F use
F userdata

sections table

F autodepth
F cct
D collected
F current
F currentblock
F currentid
F currentlevel
F currentsectionindex
F depthnumber
F finddata
F findnumber
F fullnumber
F getcurrentlevel
F getlevel
F getnumber
F internalreference
D levelmap
D levels
F matchingtilldepth
F numberatdepth
F numbers
F popblock
F prefixedconverted
F pushblock
F register
D registered
F reportstructure
F save
T sectionblockdata

T bodypart
S bookmark
F set
F setblock
F setchecker
F setentry
F setinitialblock
F setlevel
F setnumber
F showstructure
F structuredata
F title
D tobesaved
F typesetnumber
F userdata
B verbose

sets table
F define
F get
F getall
D setlist

specials table
D collected
F retrieve
F store
D tobesaved

synonyms table
F analyzed
D collected
F compare
F define
F enhance
F filter
F finalize
F flush
F isshown
F isused
F meaning
F pages
F prepare
F process
F register
F registershown
F registerused
F reset
F resetall
F resetlist
F resetshown
F resetused
F sort
F synonym
D tobesaved

tags table
T backend
T checks
F marginanchor
F margintext
F math
F mrow
F mtable
T destinationhash
T extras
F adddestination
F addimplicit
F addinternal
F addreference
F combination
F delimitedsymbol
F description
F descriptionsymbol
F document
F float
F formulacontent
F image
F item
F itemgroup
F link
F listitem
F mmultiscripts
F msub
F msubsup
F msup
F paragraph
F publication
F reference
F registerlocation
F registerpages
F registerseparator

```
    F section
    F sorting
    F subsentencesymbol
    F synonym
    F tablecell
    F tabulate
    F tabulatecell
  T finalizers
    F descriptions
  T fixes
    F linenumber
  T listdata
  T referencehash
  F setattribute
  T usedimages
    T image
  T usedstyles
    T construct
    T highlight
F current
F elementtag
F enabled
F exportbox
F finishexport
F getextradata
F getid
F getmetadata
F gettablecell
F gettabulatecell
F handler
F initializeexport
D labels
F last
F lastinchain
F localexport
F locatedtag
T patterns
  U splitter
D properties
F registerextradata
F registermetadata
F restart
F setaspect
F setattributehash
F setcodepoint
F setcombination
F setconstruct
F setfigure
F setfloat
F setformulacontent
F sethighlight
F setitem
F setitemgroup
F setlist
F setnotation
F setnotationsymbol
F setparagraph
F setproperty
F setpublication
F setregister
F setsorting
F setsynonym
F settablecell
F settabulatecell
F settagdelimitedsymbol
F settagsubsentencesymbol
F setupexport
F setuserproperties
D specifications
F start
F stop
F strip
D taglist
T usewithcare
```

table (basic lua)

allkeys	function
append	function
are_equal	function
collapsed	function
collapsedhash	function
combine	function
compact	function
concat	function
contains	function
copy	function
count	function
create	function
derive	function
deserialize	function
drop	function
fastcopy	function
fastserialize	function
filtered	function
flattened	function
fromhash	function
getmetatablekey	function
getn	function
has_one_entry	function
hashed	function
identical	function
imerge	function
imerged	function
insert	function
is_empty	function
is_simple_table	function
keys	function
load	function
loweredkeys	function
makeweak	function
merge	function
merged	function
mirrored	function
move	function
ordered	function
orderedhash	function
pack	function
prepend	function
print	function
remove	function
reverse	function
reversed	function
save	function
sequenced	function
serialize	function
setmetatablecall	function
setmetatableindex	function
setmetatableindices	function
setmetatablekey	function
setmetatablenewindex	function
sort	function
sorted	function
sortedhash	function
sortedhashkeys	function
sortedhashonly	function
sortedindexonly	function
sortedkeys	function
sortedpairs	function
sparse	function
strip	function
sub	function
swapped	function
tocontext	function
tocsv	function
tofile	function
tohandle	function
tohash	function
toxml	function
twowaymapper	function
unique	function
unnest	function
unpack	function
values	function

tex (basic tex)

amcode	table
attribute	table
automigrationcodes	data table
autoparagraphcodes	data table
badness	function
boundaries	table
D names	
D numbers	
F system	
box	table
boxresources	table
F getbox	
F getdimensions	
F save	
F use	
breakcodes	data table
buildcodes	data table
catcode	table
charactertagcodes	data table
chardef	function
classoptioncodes	data table
codes	table
conditionals	data table
constants	data table
count	table
cprint	function
ctdcatcodes	number
ctxcatcodes	number
definefont	function
delcode	table
dimen	table
dimensiondef	function
directioncodes	data table
directives	data table
discoptioncodes	data table
dontcomplain	function
doublescriptoptioncodes	data table
enableprimitives	function
error	function
expandasvalue	function
experiments	data table
extraprimitives	function
fatalerror	function
fillcodes	data table
fitnesscodes	data table
flagcodes	data table
fontidentifier	function
fontname	function
forcehmode	function
forceinputstatefile	function
forceinputstateline	function
fprint	function
frozenparcodes	data table
get	function
getalignmentcontextvalues	function
getamcode	function
getappendlinecontextvalues	function
getattribute	function
getautomigrationvalues	function
getautoparagraphvalues	function
getbox	function
getboxdir	function
getboxresourcebox	function
getboxresourcedimensions	function
getbreakcontextvalues	function
getbuildcontextvalues	function
getcardinalvalue	function
getcatcode	function
getcharactertagvalues	function
getcount	function
getdelcode	function
getdelcodes	function
getdimen	function
getdimensionvalue	function
getdirectionvalues	function
getdiscoptionvalues	function
getdiscstatevalues	function
getdoublescriptoptionvalues	function
geterrorvalues	function
getfillvalues	function
getfitnessvalues	function
getflagvalues	function

getfloat	function
getfontoffamily	function
getfrozenparvalues	function
getglue	function
getglueoptionvalues	function
getglyphdata	function
getglyphoptionvalues	function
getglyphscales	function
getglyphscript	function
getglyphstate	function
getgroupvalues	function
gethccode	function
gethelptext	function
gethmcode	function
gethyphenationvalues	function
getiftypes	function
getinputstatefile	function
getinputstateline	function
getinputstatemode	function
getinsertcontent	function
getinsertdepth	function
getinsertdistance	function
getinsertheight	function
getinsertlimit	function
getinsertmultiplier	function
getinsertwidth	function
getintegervalue	function
getinteraction	function
getinteractionmodes	function
getiovalues	function
getkerneloptionvalues	function
getlargestusedmark	function
getlccode	function
getlinedir	function
getlist	function
getlistanchorvalues	function
getlistgeometryvalues	function
getlistsignvalues	function
getlocalbox	function
getlocallevel	function
getmark	function
getmath	function
getmathclassoptionvalues	function
getmathcode	function
getmathcodes	function
getmathcontrolvalues	function
getmathdir	function
getmathgluevalues	function
getmathoptionvalues	function
getmathparametervalues	function
getmathscriptordervalues	function
getmathscriptsmodevalues	function
getmathstylenamevalues	function
getmathstylevalues	function
getmathsurroundvalues	function
getmathvariantvalues	function
getmode	function
getmodevalues	function
getmuglue	function
getmuskip	function
getnest	function
getnestlevel	function
getnoadoptionvalues	function
getnormalizelinevalues	function
getnormalizeparvalues	function
getoutputactive	function
getpacktypevalues	function
getpagecontextvalues	function
getpagestate	function
getparametermodevalues	function
getparcontextvalues	function
getpardir	function
getparmodevalues	function
getparpassclassvalues	function
getparstate	function
getpartriggervalues	function
getpenaltyoptionvalues	function
getpositvalue	function
getprepoststatevalues	function
getprimitiveorigins	function
getprotrusionboundaryvalues	function
getregisterindex	function
getrunstatevalues	function
getsfcode	function
getshapingpenaltiesvalues	function
getskip	function
getspecialmathclassvalues	function
gettextcontrolvalues	function

gettextdir function
gettoks function
getuccode function
getunitclassvalues function
glue table
glueoptioncodes data table
glyphoptioncodes data table
groupcodes data table
hashtokens function
hccode table
hmcode table
hshcatcodes number
hyphenationcodes data table
ifs data table
inicatcodes number
integerdef function
interactioncodes data table
isattribute function
isbox function
iscount function
isdefined function
isdimen function
isfloat function
isglue function
ismuglue function
ismuskip function
isprintable function
isskip function
istoks function
lastnodetype function
lccode table
linebreak function
listanchorcodes data table
listgeometrycodes data table
lists table
listsigncodes data table
luacatcodes number
magicconstants table
N assumedmathcontrol
N awfulbad
N decentcriterion
N defaultcatcodetable
N defaultdeadcycles
N defaulteqnogapstep
N defaulthangafter
N defaultoutputbox
N defaultpredisplaygap
N defaultrule
N defaultspacefactor
N defaulttolerance
N deplorable
N ejectpenalty
N ignoredepth
N infinitebad
N infinitepenalty
N infinity
N largewidthexcess
N loosecriterion
N mathallclass
N mathbeginclass
N mathdefaultpenalty
N mathendclass
N mathfirstuserclass
N mathlastuserclass
N maxattributeregisterindex
N maxboxregisterindex
N maxbytecodeindex
N maxcalculatedbadness
N maxcardinal
N maxcategorycode
N maxcharactercode
N maxdatavalue
N maxdimen
N maxdimenregisterindex
N maxdimension
N maxdimensionregisterindex
N maxendlinecharacter
N maxfloatregisterindex
N maxfontadjustshrinkfactor
N maxfontadjuststep
N maxfontadjuststretchfactor
N maxfunctionreference
N maxglueregisterindex
N maxhalfvalue
N maxhalfword
N maxinteger
N maxintegerregisterindex
N maxintregisterindex

N maxlimitedscale
N maxmarkindex
N maxmathclasscode
N maxmathfamilyindex
N maxmathscalingfactor
N maxmathstylescale
N maxmuglueregisterindex
N maxnewlinecharacter
N maxnofbytecodes
N maxnofcatcodetables
N maxnoffonts
N maxnoflanguages
N maxnofmarks
N maxnofmathfamilies
N maxquarterword
N maxscalefactor
N maxsizeofword
N maxspacefactor
N maxtoksregisterindex
N mincardinal
N mindatavalue
N mindimen
N mindimension
N minhalfword
N mininfinity
N mininteger
N minquarterword
N minscalefactor
N minspacefactor
N nocatcodetable
N null
N nullflag
N nullfont
N onebp
N presetrulethickness
N runningrule
N smallstretchability
N texeqtbsize
N texhashprime
N texhashsize
N two
N undefinedmathparameter
N unity
N unusedattributevalue
N unusedmathfamily
N unusedmathstyle
N unusedscriptvalue
N unusedstatevalue
N zeroglue

mathchardef	function
mathcode	table
mathcontrolcodes	data table
mathoptioncodes	data table
mathparametercodes	data table
mathscriptordercodes	data table
mathscriptsmodecodes	data table
modelevels	table
N horizontal	
N math	
N unset	
N vertical	
modes	data table
mprint	function
muglue	table
muskip	table
nest	table
nilcatcodes	number
noadoptioncodes	data table
normalizelinecodes	data table
normalizeparcodes	data table
notcatcodes	number
number	function
parcontextcodes	data table
parpassclasscodes	data table
partriggercodes	data table
penaltyoptioncodes	data table
poplocal	function
popsavelevel	function
posit	table
positdef	function
preparelinebreak	function
primitives	function
print	function
protrusionboundarycodes	data table
prtcatcodes	number
pushlocal	function
pushsavelevel	function
quitlocal	function

quittoks	function
resetparagraph	function
rlncatcodes	number
romannumeral	function
round	function
runlocal	function
runstring	function
runtoks	function
saveboxresource	function
scale	function
scantoks	function
set	function
setamcode	function
setattribute	function
setbox	function
setboxdir	function
setcardinalvalue	function
setcatcode	function
setcount	function
setdelcode	function
setdimen	function
setdimensionvalue	function
setfloat	function
setglue	function
setglyphdata	function
setglyphscript	function
setglyphstate	function
sethccode	function
sethmcode	function
setinputstatefile	function
setinputstateline	function
setinputstatemode	function
setinsertcontent	function
setinsertdistance	function
setinsertlimit	function
setinsertmultiplier	function
setintegervalue	function
setinteraction	function
setlccode	function
setlinedir	function
setlist	function
setlocalbox	function
setmath	function
setmathcode	function
setmathdir	function
setmuglue	function
setmuskip	function
setnest	function
setpardir	function
setpositvalue	function
setrunstate	overloaded functi
setsfcode	function
setskip	function
settextdir	function
settoks	function
setuccode	function
sfcode	table
shapingpenaltiescodes	data table
shiftparshape	function
shipout	function
showcontext	function
skip	table
snapshotpar	function
sp	function
specialmathclasscodes	data table
splitbox	function
sprint	function
stringtocodesbitmap	function
systemmodes	data table
takebox	function
texcatcodes	number
textcontrolcodes	data table
toks	table
tonumber	function
toscaled	function
tpacatcodes	number
tpbcatcodes	number
tprint	function
trackers	data table
triggerbuildpage	function
txtcatcodes	number
uccode	table
uprint	function
useboxresource	function
userdiscoptioncodes	data table
userglyphoptioncodes	data table
vrbcatcodes	number
write	function

texconfig (basic tex)

buffersize table
- N size
- N step

errorlinesize table
- N size

expandsize table
- N size

filesize table
- N size
- N step

firstline string

fontsize table
- N size
- N step

formatname string

functionsize number

halferrorlinesize table
- N size

hashsize table
- N size
- N step

init overloaded function

inputsize table
- N size
- N step

insertsize table
- N size
- N step

kpse_init boolean

languagesize table
- N size
- N step

marksize table
- N size
- N step

nestsize table
- N size
- N step

nodesize table
- N size
- N step

parametersize table
- N size
- N step

poolsize table
- N size
- N step

propertiessize number

savesize table
- N size
- N step

shell_escape string

stringsize table
- N size
- N step

tokensize table
- N size
- N step

texio (basic tex)

closeinput	function	**write**	function
forceendoffile	function	**write_nl**	function
getinputindex	function	**writenl**	function
getsourcefilename	function	**writeselector**	function
setlogfile	function	**writeselectorlf**	function
terminal	function	**writeselectornl**	function

token (basic tex)

create	function	getdimension	function
expand	function	getexpandable	function
expand_macro	function	getexpansion	function
expandmacro	function	getfields	function
future_expand	function	getflags	function
futureexpand	function	getfrozen	function
get_active	function	getfunctionvalues	function
get_cmdchrcs	function	getid	function
get_cmdname	function	getimmutable	function
get_command	function	getindex	function
get_csname	function	getinstance	function
get_cstoken	function	getinteger	function
get_dimension	function	getmacro	function
get_expandable	function	getmeaning	function
get_fields	function	getnoaligned	function
get_flags	function	getparameters	function
get_frozen	function	getpermanent	function
get_id	function	getprimitive	function
get_immutable	function	getprimitives	function
get_index	function	getprotected	function
get_instance	function	getrange	function
get_integer	function	gettok	function
get_macro	function	gettolerant	function
get_meaning	function	gobble	function
get_noaligned	function	gobble_dimen	function
get_parameters	function	gobble_integer	function
get_permanent	function	gobbledimen	function
get_primitive	function	gobbleinteger	function
get_protected	function	grab	function
get_range	function	is_defined	function
get_tok	function	is_next_char	function
get_tolerant	function	is_token	function
getactive	function	isdefined	function
getcmdchrcs	function	isnextchar	function
getcmdname	function	istoken	function
getcommand	function	new	function
getcommandid	function	peek_next	function
getcommandvalues	function	peek_next_char	function
getconstant	function	peek_next_expanded	function
getcsname	function	peeknext	function
getcstoken	function	peeknextchar	function

peeknextexpanded	function
pop_macro	function
popmacro	function
push_macro	function
pushmacro	function
put_back	function
put_next	function
putback	function
putnext	function
save_lua	function
savelua	function
scan_argument	function
scan_box	function
scan_cardinal	function
scan_char	function
scan_cmdchr	function
scan_cmdchr_expanded	function
scan_code	function
scan_csname	function
scan_delimited	function
scan_dimen	function
scan_dimen_argument	function
scan_float	function
scan_glue	function
scan_integer	function
scan_integer_argument	function
scan_key	function
scan_keyword	function
scan_keyword_cs	function
scan_letters	function
scan_luacardinal	function
scan_luainteger	function
scan_luanumber	function
scan_next	function
scan_next_char	function
scan_next_expanded	function
scan_real	function
scan_scale	function
scan_skip	function
scan_string	function
scan_token	function
scan_token_code	function
scan_tokenlist	function
scan_toks	function
scan_value	function
scan_word	function
scanargument	function
scanbox	function
scancardinal	function
scanchar	function
scancmdchr	function
scancmdchrexpanded	function
scancode	function
scancsname	function
scancstoken	function
scandelimited	function
scandetokened	function
scandimen	function
scandimenargument	function
scandimension	function
scandimensionargument	function
scanfloat	function
scanglue	function
scanint	function
scaninteger	function
scanintegerargument	function
scankey	function
scankeyword	function
scankeywordcs	function
scanletters	function
scanluacardinal	function
scanluainteger	function
scanluanumber	function
scannext	function
scannextchar	function
scannextexpanded	function
scanposit	function
scanreal	function
scanscale	function
scanskip	function
scanstring	function
scantoken	function
scantokencode	function
scantokenlist	function
scantokenstring	function
scantoks	function
scanvalue	function
scanword	function

serialize	function	**setlua**	function
set_char	function	**setmacro**	function
set_dimension	function	**skip_next**	function
set_integer	function	**skip_next_expanded**	function
set_lua	function	**skipnext**	function
set_macro	function	**skipnextexpanded**	function
setchar	function	**type**	function
setdimension	function	**undefine_macro**	function
setinteger	function	**undefinemacro**	function

tokens (context)

accessors table
- F active
- F cmd
- F cmdchrcs
- F cmdname
- F command
- F csname
- F expandable
- F frozen
- F index
- F name
- F protected
- F range

astable function

bits table
- N active
- N alignment
- N begingroup
- N character
- N close
- N comment
- N endgroup
- N endofline
- N escape
- N ignore
- N invalid
- N letter
- N mathshift
- N open
- N other
- N parameter
- N space
- N subscript
- N superscript
- N whitespace

cache data table

commands data table

compile function

constant function

converters table
- S toboolean
- S todimen
- S toglue
- S tonumber
- S tostring

create function

defined function

flags function

getinteger function

getters table
- F box
- F count
- F cstoken
- F dimen
- F glue
- F macro
- F meaning
- F skip
- F token

gobble function

isdefined function

istoken function

new function

pickup function

popmacro function

presets table
- T 1 argument
- T 1 integer
- T 1 optional
- T 1 string
- T 2 arguments
- T 2 integers
- T 2 optionals
- T 2 strings
- T 3 arguments
- T 3 integers
- T 3 optionals
- T 3 strings
- T 4 arguments
- T 4 integers
- T 4 optionals
- T 4 strings

T 5 strings
T 6 strings
T 7 strings
T 8 strings

pushmacro function

scanners table

F argument
F argumentasis
F array
F boolean
F box
F bracketed
F bracketedasis
F cardinal
F char
F cmdchr
F cmdchrexpanded
F code
F conditional
F count
F csname
F csnameunchecked
F delimited
F detokened
F dimen
F dimenargument
F dimension
F float
F glue
F gluespec
F gluevalues
F hash
F hbox
F integer
F integerargument
F ischar
F key
F keyword
F keywordcs
F letters
F list
F luacardinal
F luainteger
F luanumber
F lxmlid
F next
F nextchar
F nextexpanded
F number
F optional
F peek
F peekchar
F peekexpanded
F posit
F real
F scanclose
F scanopen
F skip
F skipexpanded
F string
F table
F token
F tokencode
F tokenlist
F tokens
F tokenstring
F toks
F value
F vbox
F verbatim
F vtop
F whd
F word

setinteger function

setters table

F box
F char
F count
F dimen
F glue
F lua
F macro
F skip

shortcuts table

T bits
N active
N alignment
N begingroup

```
  N character
  N close
  N comment
  N endgroup
  N endofline
  N escape
  N ignore
  N invalid
  N letter
  N mathshift
  N open
  N other
  N parameter
  N space
  N subscript
  N superscript
  N whitespace
N close
F inspect
N open
F report
F scanargument
F scanargumentasis
F scanarray
F scanboolean
F scanbox
F scanbracketed
F scanbracketedasis
F scancardinal
F scanclose
F scancode
F scanconditional
F scancsname
F scancsnameunchecked
F scandelimited
F scandetokened
F scandimen
F scandimenargument
F scandimension
F scanfloat
F scanglue
F scangluespec
F scangluevalues
F scanhash
F scanhbox
F scaninteger
F scanintegerargument
F scankeyword
F scankeywordcs
F scanletters
F scanlist
T scanners
  F argument
  F argumentasis
  F array
  F boolean
  F box
  F bracketed
  F bracketedasis
  F cardinal
  F char
  F cmdchr
  F cmdchrexpanded
  F code
  F conditional
  F count
  F csname
  F csnameunchecked
  F delimited
  F detokened
  F dimen
  F dimenargument
  F dimension
  F float
  F glue
  F gluespec
  F gluevalues
  F hash
  F hbox
  F integer
  F integerargument
  F ischar
  F key
  F keyword
  F keywordcs
  F letters
  F list
  F luacardinal
```

- F luainteger
- F luanumber
- F lxmlid
- F next
- F nextchar
- F nextexpanded
- F number
- F optional
- F peek
- F peekchar
- F peekexpanded
- F posit
- F real
- F scanclose
- F scanopen
- F skip
- F skipexpanded
- F string
- F table
- F token
- F tokencode
- F tokenlist
- F tokens
- F tokenstring
- F toks
- F value
- F vbox
- F verbatim
- F vtop
- F whd
- F word
- F scannumber
- F scanopen
- F scanoptional
- F scanstring
- F scantable
- F scantokenlist
- F scantokenstring
- F scantoks
- F scanvbox
- F scanverbatim
- F scanvtop
- F scanword
- F toboolean
- F todimen
- T tokens
  - T accessors
    - F active
    - F cmd
    - F cmdchrcs
    - F cmdname
    - F command
    - F csname
    - F expandable
    - F frozen
    - F index
    - F name
    - F protected
    - F range
  - F astable
  - D cache
  - D commands
  - F compile
  - F constant
  - T converters
    - S toboolean
    - S todimen
    - S toglue
    - S tonumber
    - S tostring
  - F create
  - F defined
  - F flags
  - F getinteger
  - T getters
    - F box
    - F count
    - F cstoken
    - F dimen
    - F glue
    - F macro
    - F meaning
    - F skip
    - F token
  - F gobble
  - F isdefined
  - F istoken
  - F new

F pickup
F popmacro
T presets
F pushmacro
F setinteger
T setters
  F box
  F char
  F count
  F dimen
  F glue
  F lua
  F macro
  F skip
T shortcuts
  N close
  F inspect
  N open
  F report
  F scanargument
  F scanargumentasis
  F scanarray
  F scanboolean
  F scanbox
  F scanbracketed
  F scanbracketedasis
  F scancardinal
  F scanclose
  F scancode
  F scanconditional
  F scancsname
  F scancsnameunchecked
  F scandelimited
  F scandetokened
  F scandimen
  F scandimenargument
  F scandimension
  F scanfloat
  F scanglue
  F scangluespec
  F scangluevalues
  F scanhash
  F scanhbox
  F scaninteger
  F scanintegerargument
  F scankeyword
  F scankeywordcs
  F scanletters
  F scanlist
  F scannumber
  F scanopen
  F scanoptional
  F scanstring
  F scantable
  F scantokenlist
  F scantokenstring
  F scantoks
  F scanvbox
  F scanverbatim
  F scanvtop
  F scanword
  F toboolean
  F todimen
  F tonumber
  F tostring
 D values
F tonumber
F tostring

values data table

tracers (context)

printerror function
printwarning function
showlines function

trackers (context)

data	data table	**register**	function
default	function	**report**	function
disable	function	**reset**	function
done	table	**savehash**	function
dumphashtofile	function	**show**	function
enable	function	**value**	function
list	function	**values**	function
name	string		

typesetters (context)

alignments table

breakpoints table

F define

F handler

F last

D mapping

T methods

T numbers

F set

F setreplacement

F show

cases table

T actions

F handler

F register

F set

characteralign table

F handler

F resetcharacteralign

F setcharacteralign

characters table

F handler

T methods

F replacenbsp

F replacenbspaces

checkers table

F handler

F reset

F set

cleaners table

F handler

F set

corrections table

F handler

digits table

T actions

F handler

F set

directions table

F getfences

F getglobal

F getmethod

F handler

T handlers

F installhandler

F set

F setcolor

F tomode

firstlines table

T actions

F default

F line

F word

F handler

F set

fontkerns table

F handler

initials table

T actions

F default

F handler

F set

injectors table

F check

T list

F mark

F reset

F set

F show

italics table

F enablemath

F enabletext

F handler

kerns table

D factors

F handler

F keepligature

D mapping

F set

localboxes table

F handler

manipulators table

F apply
F applyspecification
T methods
F Word
F Words
F abbreviatedjournal
F expandedjournal
F lowercase
F stripperiod
F uppercase
T patterns
U list
U pair
F splitspecification

margins table
F finalhandler
F globalhandler
F localhandler
F save

marksuspects function

paragraphs table
F addanchortoline
F addtoline
F calculatedelta
F checkline
F moveinline

periodkerns table
D factors
F handler
D mapping
F set

processors table
F apply
F register
F reset
F split
F startapply
F stopapply
F stripped
F tostring

rubies table
F attach
F check

showsuspects function

spacings table
F define
F handler
D mapping
T numbers
T frenchpunctuation
T characters
S name
N number
F reset
F set
F setup

synchronize table
F handler

tighten table
F handler

wrappers table

xtables table
F cleanup
F construct
F create
F finish_row
F flush
F initialize_construct
F initialize_reflow_height
F initialize_reflow_width
F next_row
F reflow_height
F reflow_width
F set_construct
F set_reflow_height
F set_reflow_width

updaters (context)

apply function
register function

url (context)

addscheme	function	**hashed**	function
barepath	function	**hasscheme**	function
construct	function	**query**	function
decode	function	**split**	function
encode	function	**toquery**	function
escape	function	**unescape**	function
filename	function	**unescapeget**	function

utf8 (basic lua)

char	function	**codes**	function
charpattern	string	**len**	function
codepoint	function	**offset**	function

utilities (context)

debugger table

- F disable
- F enable
- F savestats
- F showstats
- F showtraceback

files table

- F close
- F getposition
- F getsize
- F look
- F open
- F read2dot14
- F readbyte
- F readbytes
- F readbytetable
- F readcardinal
- F readcardinal1
- F readcardinal1le
- F readcardinal2
- F readcardinal2le
- F readcardinal3
- F readcardinal3le
- F readcardinal4
- F readcardinal4le
- F readcardinaltable
- F readchar
- F readcline
- F readcstring
- F readfixed2
- F readfixed4
- F readinteger
- F readinteger1
- F readinteger1le
- F readinteger2
- F readinteger2le
- F readinteger3
- F readinteger3le
- F readinteger4
- F readinteger4le
- F readintegertable
- F readsignedbyte
- F readstring
- F readtag
- F setposition
- F size
- F skip
- F skiplong
- F skipposition
- F skipshort
- F writebyte
- F writecardinal1
- F writecardinal1le
- F writecardinal2
- F writecardinal2le
- F writecardinal3
- F writecardinal3le
- F writecardinal4
- F writecardinal4le
- F writeinteger1
- F writeinteger1le
- F writeinteger2
- F writeinteger2le
- F writeinteger3
- F writeinteger3le
- F writeinteger4
- F writeinteger4le
- F writestring

formatters table

- F formatcolumns
- F list
- F stripzeros

ghostscript table

- F convert
- F statistics

graphicmagick table

- F convert
- F statistics

graphicsmagick table

- F convert
- F statistics

lua table

B alwaysstripcode
F checkmemory
F compile
F loadedluacode
F loadstring
F loadstripped
N nofstrippedbytes
N nofstrippedchunks
F registerfinalizer
B stripcode
T strippedchunks
F strippedloadstring
T suffixes
S lmt
S lua
S luc
S lui
S luj
S luv
S tma
S tmc
S tua
S tuc

packers table

F hashed
F new
F pack
F simplehashed
F strip
F unpack
N version

parsers table

F add_settings_to_array
F arguments_to_table
F array_to_string
F csvsplitter
F getparameters
F groupedsplitat
F hash_to_string
T hashes
T settings_to_list
T tablet
T settings_to_set
T min,max,fit,broad,auto
B auto
B broad
B fit
B max
B min
T text,hidden,paper,page,leftpage,rightpage
B hidden
B leftpage
B page
B paper
B rightpage
B text
F iterator
F keq_to_hash
F listitem
F make_settings_to_hash_pattern
F mergehashes
F options_to_array
F options_to_hash
T patterns
U settings_to_array
U settings_to_hash_a
U settings_to_hash_b
U settings_to_hash_c
U settings_to_hash_d
U unittotex
F process_stripped_settings
F rfc4180splitter
F runtime
F settings_to_array
F settings_to_array_obey_fences
F settings_to_array_with_repeat
F settings_to_hash
F settings_to_hash_colon_too
F settings_to_hash_strict
F settings_to_hash_tolerant
F settings_to_numbers
F settings_to_set
F simple_hash_to_string
F splitmethod
F splitthousands
F stepper
F stripped_settings_to_array
F totime

F unittotex
F unittoxml

randomizer table

F get
F getnumber
F getrepeatable
F getrepeatableseed
F getseed
F newrepeatable
F popseed
F pushseed
F repeatable
F reuseseed
F setseed
F setseedi

randomizers table

F poisson

sequencers table

F appendaction
F appendgroup
F compile
F disableaction
F disablegroup
F enableaction
F enablegroup
D functions
F localize
F new
F nodeprocessor
F prependaction
F prependgroup
F removeaction
F replaceaction
F setkind
F tostring

setters table

F disable
F enable
F initialize
F list
F new
F register
F report
F reset
F show

stacker table

F new

storage table

F allocate
F checked
F mark
F marked
F setinitializer
F sparse

streams table

F close
F getposition
F getsize
F getstring
F look
F open
F openstring
F read2dot14
F readbyte
F readbytes
F readbytetable
F readcardinal
F readcardinal1
F readcardinal1le
F readcardinal2
F readcardinal2le
F readcardinal3
F readcardinal3le
F readcardinal4
F readcardinal4le
F readcardinaltable
F readchar
F readcline
F readcstring
F readfixed2
F readfixed4
F readinteger
F readinteger1
F readinteger1le
F readinteger2
F readinteger2le
F readinteger3
F readinteger3le

F readinteger4
F readinteger4le
F readintegertable
F readsignedbyte
F readstring
F setposition
F size
F skip
F skipbytes
F skiplong
F skipshort
F tocardinal1
F tocardinal1le
F tocardinal2
F tocardinal2le
F tocardinal3
F tocardinal3le
F tocardinal4
F tocardinal4le

strings table

F collapse
F collapsecrlf
T formatters
  F add
  F new
F newcollector
F newrepeater
F nice
T striplinepatterns
  U collapse
  U collapse all
  U prune
  U prune and collapse
  U prune and no empty
  U prune and to space
  U retain
  U retain and collapse
  U retain and no empty
F striplines
F striplong
F tabtospace
T tracers

tables table

F accesstable
F definedtable
F definetable
F encapsulate
F insertaftervalue
F insertbeforevalue
F migratetable
F removevalue
F replacevalue

tar table

T archives
F closearchive
T files
  F fetch
  F hash
  F list
  F pack
  F unpack
F openarchive
T registeredfiles
T streams
  F fetch
  F hash
  F list
  F pack
  F unpack
T strings
  F fetch
  F hash
  F list
  F pack
  F unpack

templates table

F load
F replace
F replacer
F resolve

zipfiles table

F close
F found
F hash
F list
F open
F unzip

```
F unzipdir

F zipdir
```

visualizers (context)

escapepatterns	data table	**signal**	string
getspecification	function	**signalpattern**	userdata
getvisualizer	function	**specifications**	data table
load	function	**visualize**	function
makenested	function	**visualizebuffer**	function
makepattern	function	**visualizefile**	function
newgrammar	function	**visualizestring**	function
newhandler	function	**write**	function
parsedxml	function	**writeargument**	function
pattern	function	**writedefault**	function
register	function	**writeemptyline**	function
registerescapecommand	function	**writenewline**	function
registerescapeline	function	**writespace**	function
registerescapepattern	function		

xcomplex (extra lua)

```
I          function
__add      function
__div      function
__eq       function
__index    table
  F I
  F __add
  F __div
  F __eq
  T __index
    F I
    F __add
    F __div
    F __eq
    F __mul
    S __name
    F __pow
    F __sub
    F __tostring
    F __unm
    F abs
    F acos
    F acosh
    F arg
    F asin
    F asinh
    F atan
    F atanh
    F conj
    F cos
    F cosh
    F exp
    F i
    F imag
    F log
    F new
    F pow
    F proj
    F real
    F sin
    F sinh
    F sqrt
    F tan
    F tanh
    F topair
    F tostring
    F totable
  F __mul
  S __name
  F __pow
  F __sub
  F __tostring
  F __unm
  F abs
  F acos
  F acosh
  F arg
  F asin
  F asinh
  F atan
  F atanh
  F conj
  F cos
  F cosh
  F exp
  F i
  F imag
  F log
  F new
  F pow
  F proj
  F real
  F sin
  F sinh
  F sqrt
  F tan
  F tanh
  F topair
  F tostring
  F totable
__mul      function
__name     string
```

__pow	function
__sub	function
__tostring	function
__unm	function
abs	function
acos	function
acosh	function
arg	function
asin	function
asinh	function
atan	function
atanh	function
conj	function
cos	function
cosh	function
exp	function
i	function
imag	function
log	function
new	function
pow	function
proj	function
real	function
sin	function
sinh	function
sqrt	function
tan	function
tanh	function
topair	function
tostring	function
totable	function

xdecimal (extra lua)

__add function
__band function
__bor function
__bxor function
__div function
__eq function
__idiv function
__index table

```
  F __add
  F __band
  F __bor
  F __bxor
  F __div
  F __eq
  F __idiv
  T __index
    F __add
    F __band
    F __bor
    F __bxor
    F __div
    F __eq
    F __idiv
    F __le
    F __lt
    F __mod
    F __mul
    S __name
    F __pow
    F __shl
    F __shr
    F __sub
    F __tostring
    F __unm
    F abs
    F band
    F bor
    F bxor
    F conj
    F copy
    F exp
    F getprecision
    F ln
    F log
    F max
    F min
    F minus
    F new
    F plus
    F pow
    F rotate
    F setprecision
    F shift
    F sqrt
    F toengstring
    F tonumber
    F tostring
    F trim
  F __le
  F __lt
  F __mod
  F __mul
  S __name
  F __pow
  F __shl
  F __shr
  F __sub
  F __tostring
  F __unm
  F abs
  F band
  F bor
  F bxor
  F conj
  F copy
  F exp
  F getprecision
  F ln
  F log
  F max
  F min
  F minus
```

```
  F new
  F plus
  F pow
  F rotate
  F setprecision
  F shift
  F sqrt
  F toengstring
  F tonumber
  F tostring
  F trim
__le          function
__lt          function
__mod         function
__mul         function
__name        string
__pow         function
__shl         function
__shr         function
__sub         function
__tostring    function
__unm         function
abs           function
band          function
bor           function
bxor          function
conj          function
copy          function
exp           function
getprecision  function
ln            function
log           function
max           function
min           function
minus         function
new           function
plus          function
pow           function
rotate        function
setprecision  function
shift         function
sqrt          function
toengstring   function
tonumber      function
tostring      function
trim          function
```

xmath (extra lua)

acos	function
acosh	function
asin	function
asinh	function
atan	function
atan2	function
atanh	function
cbrt	function
ceil	function
copysign	function
cos	function
cosh	function
deg	function
erf	function
erfc	function
exp	function
exp2	function
expm1	function
fabs	function
fdim	function
floor	function
fma	function
fmax	function
fmin	function
fmod	function
frexp	function
gamma	function
hypot	function
inf	number
isfinite	function
isinf	function
isnan	function
isnormal	function
j0	function
j1	function
jn	function
ldexp	function
lgamma	function
log	function
log10	function
log1p	function
log2	function
logb	function
modf	function
nan	number
nearbyint	function
nextafter	function
pi	number
pow	function
rad	function
remainder	function
remquo	function
round	function
scalbn	function
sin	function
sinh	function
sqrt	function
tan	function
tanh	function
tgamma	function
trunc	function
y0	function
y1	function
yn	function

xml (context)

addentitiesdoctype function
addspecialcode function
after function
all function
applylpath function
applyselector function
assign function
attribute function
badinclusions function
before function
body function
builtinentities table
- B amp
- B apos
- B gt
- B lt
- B quot

cdata function
cdatatotext function
checkbom function
checkedempty function
checkerror function
checknamespace function
checkns function
cleansed function
collect function
~~**collect_elements**~~ ~~function~~
collect_tags function
~~**collect_texts**~~ ~~function~~
collected function
collecttexts function
compactelement function
content function
convert function
copy function
count function
cprint function
css table
- F applyselector
- F colorspecification
- F dimension
- F family
- F fontspecification
- F mappedstylevalue
- F padding
- F size
- F style
- F stylevalue
- F todimension
- F variant
- F weight

defaulthandlers table
- D file
- D parsedxml
- D string
- D tex
- D texline
- D texspace
- D verbose

defaultprotocol string
delete function
~~**delete_element**~~ ~~function~~
despecialized function
each function
~~**each_element**~~ ~~function~~
elements function
elementtocdata function
empty function
entities data table
erase function
errorhandler function
escaped function
expand function
expressions table
- F attribute
- F boolean
- F child
- F contains
- F count
- F error
- F find
- F hasclass

F idstring
F lower
F name
F number
F oneof
F print
F quit
F tag
F text
F undefined
F upper

fillin function

filter function

finalizers table

F paragraphs
T tex
F all
F attribute
F chainattribute
F chainpath
F collapsed
F command
F concat
F concatrange
F context
F count
F ctxtext
F default
F depth
F first
F function
F index
F last
F lettered
F lower
F lowerall
F match
F name
F param
F parameter
F position
F reverse
F stripped
F tag
F tags
F text
F upper
F upperall
F url
F xml
T xml
F all
F att
F attribute
F attributes
F cdata
F chainattribute
F count
F default
F elements
F empty
F first
F function
F index
F last
F lowerall
F match
F name
F position
F reverse
F table
F tag
F tags
F text
F texts
F tomacro
F upperall
F url

first function

found function

functions table

F classes
F hasclass

getbuffer function

gethandlers function

gethandlersfunction function

helpers table

F recursetext

include	function
inclusion	function
inclusions	function
index	function
inheritattributes	function
inheritedconvert	function
inject	function
~~**inject_element**~~	~~function~~
~~**inject_element_after**~~	~~function~~
~~**inject_element_before**~~	~~function~~
injectafter	function
injectbefore	function
insert	function
insert_element	function
~~**insert_element_after**~~	~~function~~
~~**insert_element_before**~~	~~function~~
insertafter	function
insertbefore	function
insertcdata	function
insertcomment	function
inspect	function
is_valid	function
kind	function
last	function
lastmatch	function
load	function
lpath	function
lpathcached	function
lpathcalls	function
lshow	function
makestandalone	function
match	function
mml	table
F stripped	
F verbose	
name	function
newhandlers	function
nodesettostring	function
originalload	function
package	function
parent	function
parsedentitylpeg	userdata
pihandlers	table
F injector	
F tex	
placeholders	table
F unknown_any_entity	
F unknown_dec_entity	
F unknown_hex_entity	
popmatch	function
position	function
privatecodes	data table
privatetoken	function
process	function
~~**process_attributes**~~	~~function~~
~~**process_elements**~~	~~function~~
processattributes	function
profiled	table
pure	function
pushmatch	function
raw	function
registerentity	function
registerns	function
reindex	function
~~**remap_name**~~	~~function~~
~~**remap_namespace**~~	~~function~~
~~**remap_tag**~~	~~function~~
remapname	function
remapnamespace	function
remaptag	function
rename	function
renamespace	function
reparsedentitylpeg	userdata
replace	function
~~**replace_element**~~	~~function~~
resetinjectors	function
resolvens	function
root	function
save	function
selection	function
separate	function
serialize	function
serializetotext	function
setcdata	function
setcomment	function
sethandlersfunction	function
setinjectors	function
setproperty	function

settings	function	**tostring**	function
specialcodes	data table	**totable**	function
specialhandler	table	**toxml**	function
sprint	function	**tprint**	function
string	function	**traverse**	function
strip	function	**unescaped**	function
~~**strip_whitespace**~~	~~function~~	**unescapedentitylpeg**	userdata
stripelement	function	**unprivatized**	function
stripleadingspaces	function	**unspecialized**	function
table	function	**wipe**	function
text	function	**withelement**	function
textonly	function	**withelements**	function
texttocdata	function	**wrap**	function
tocdata	function	**xmlns**	table
toelement	function		

xzip (extra lua)

adler32	function	**crc32**	function
compress	function	**decompress**	function
compresssize	function	**decompresssize**	function

zip (context)

archives	table	**openarchive**	function
closearchive	function	**registeredfiles**	table

zlib (context)

adler32	function	**crc32**	function
compress	function	**decompress**	function
compresssize	function	**decompresssize**	function